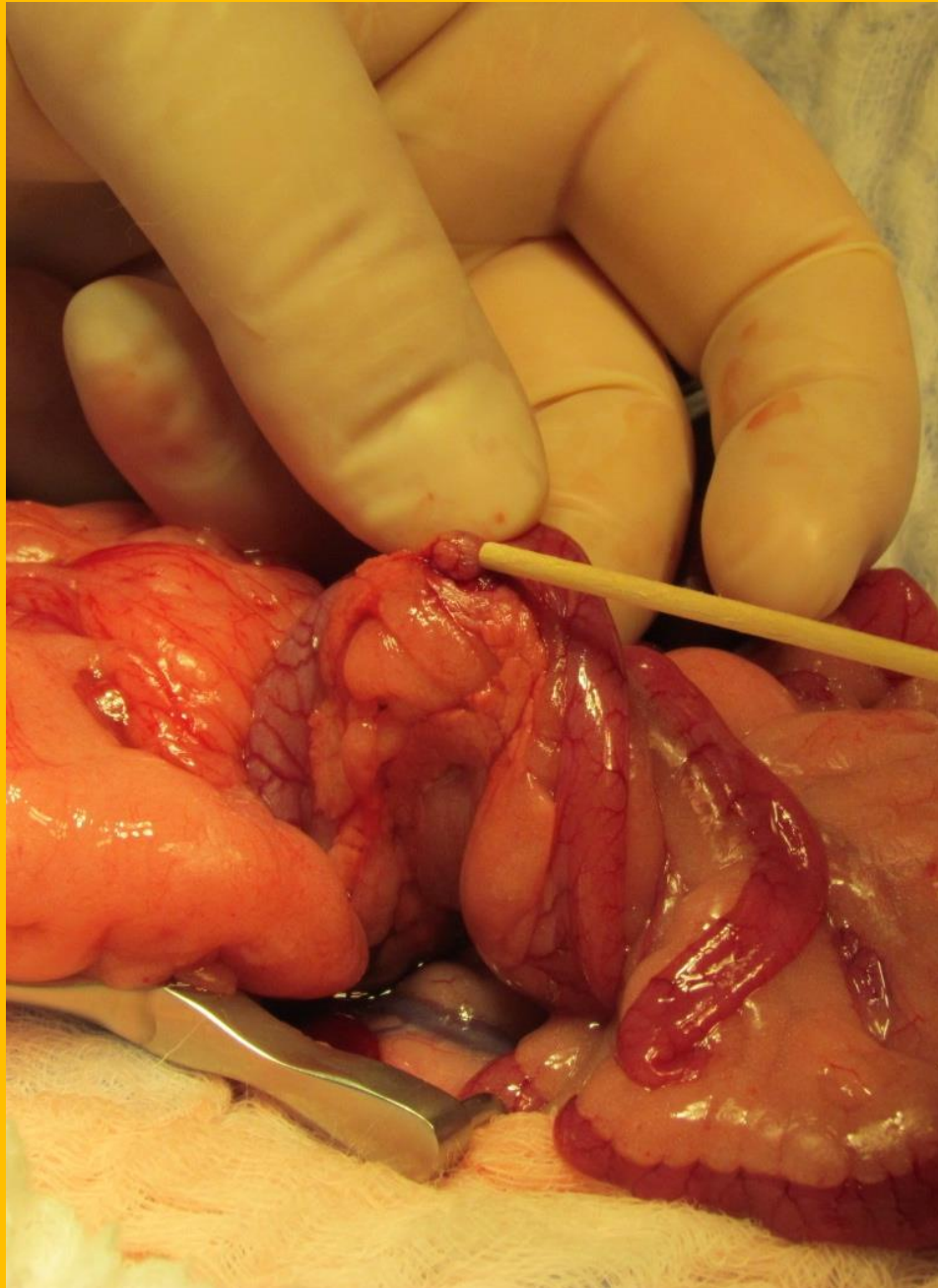




Small Exotic Mammal Soft Tissue Surgery

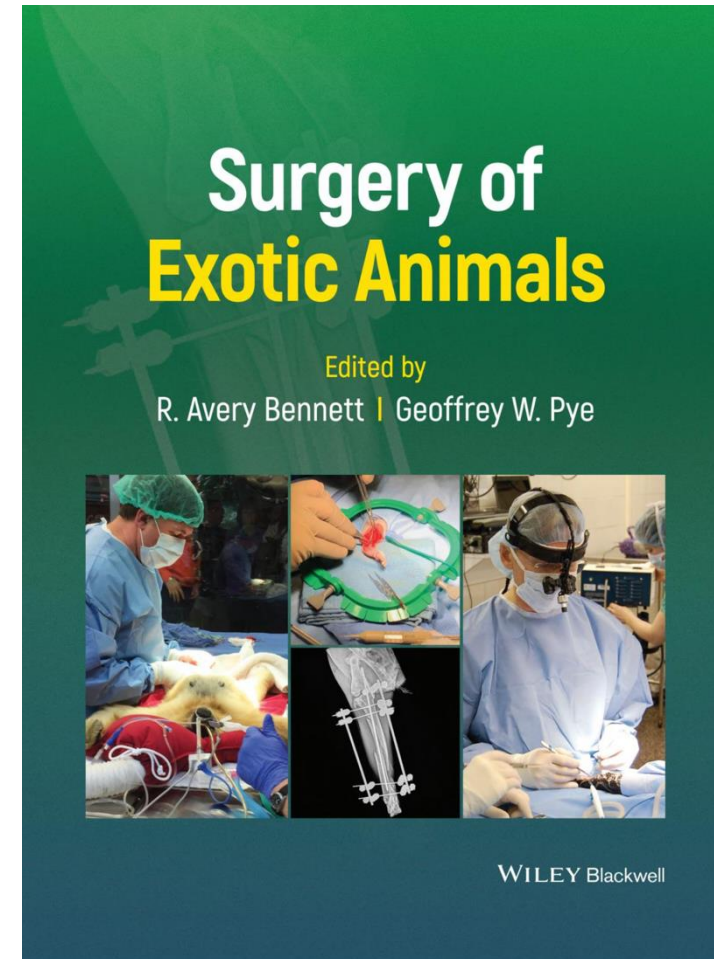
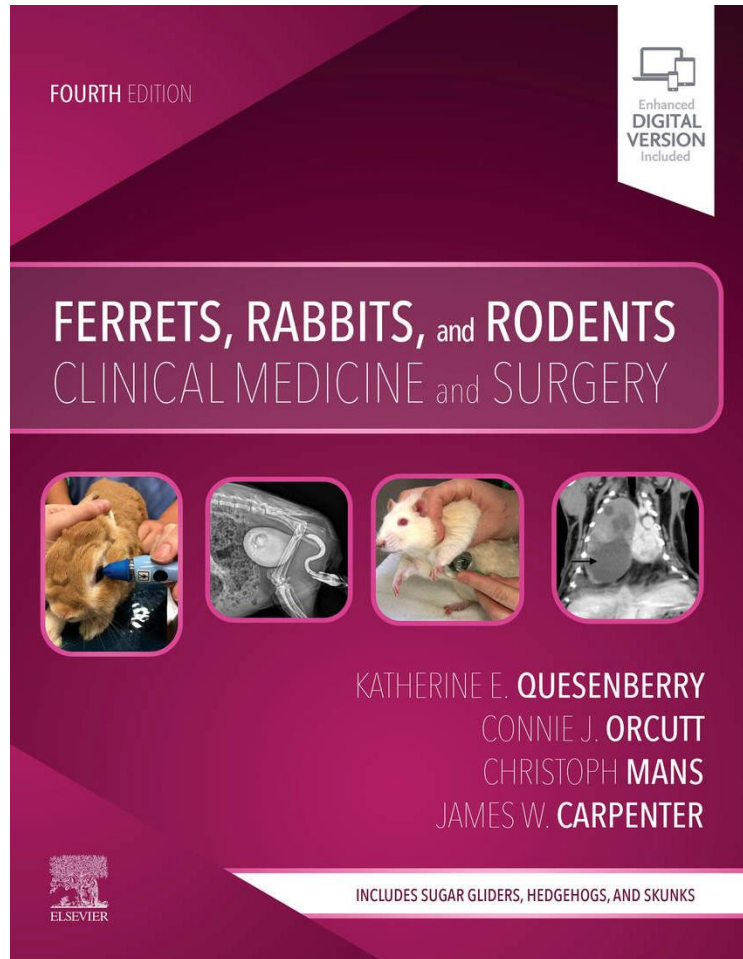
David Sanchez-Migallon Guzman, L.V., M.S.,
Dipl. ECZM(Avian, Small Mammal), Dipl. ACZM



Learning Objectives

- Be aware of the instrumentation considerations for small exotic mammals
- Know the surgical techniques for selected small exotic mammal surgical procedures

References

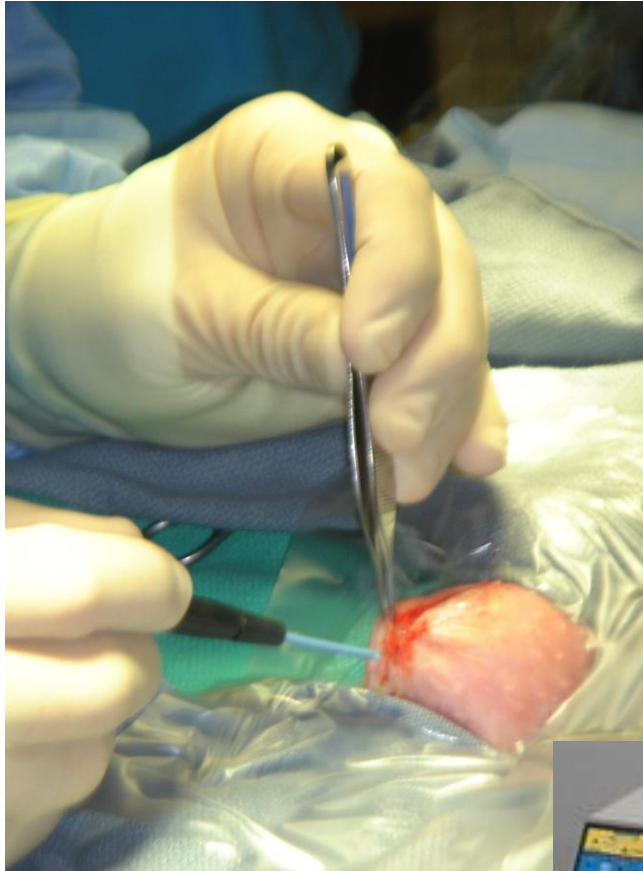


Microsurgical Instruments

- Standard length
- Rounded handles
- Miniaturized ends
- Counterbalanced
- Hands rest on table



Radiosurgery

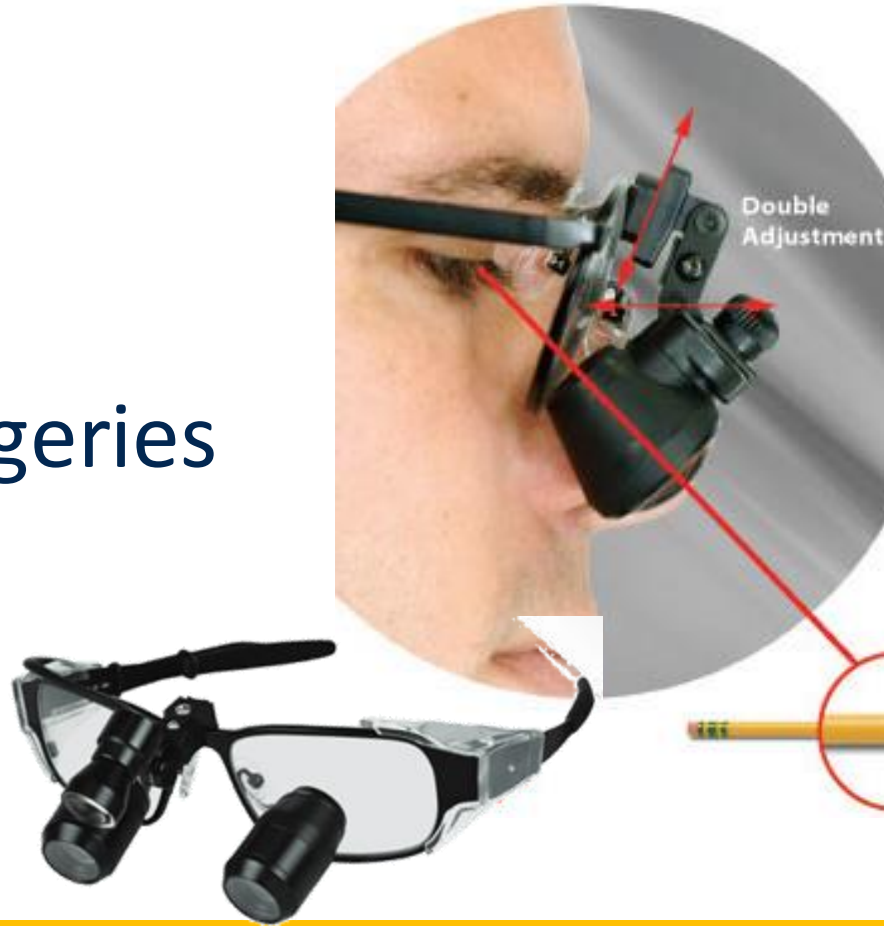


- 3.8 - 4.0 MHz
- Antenna plate
- Flammable liquids
- Cut and coagulate
- Monopolar, bipolar

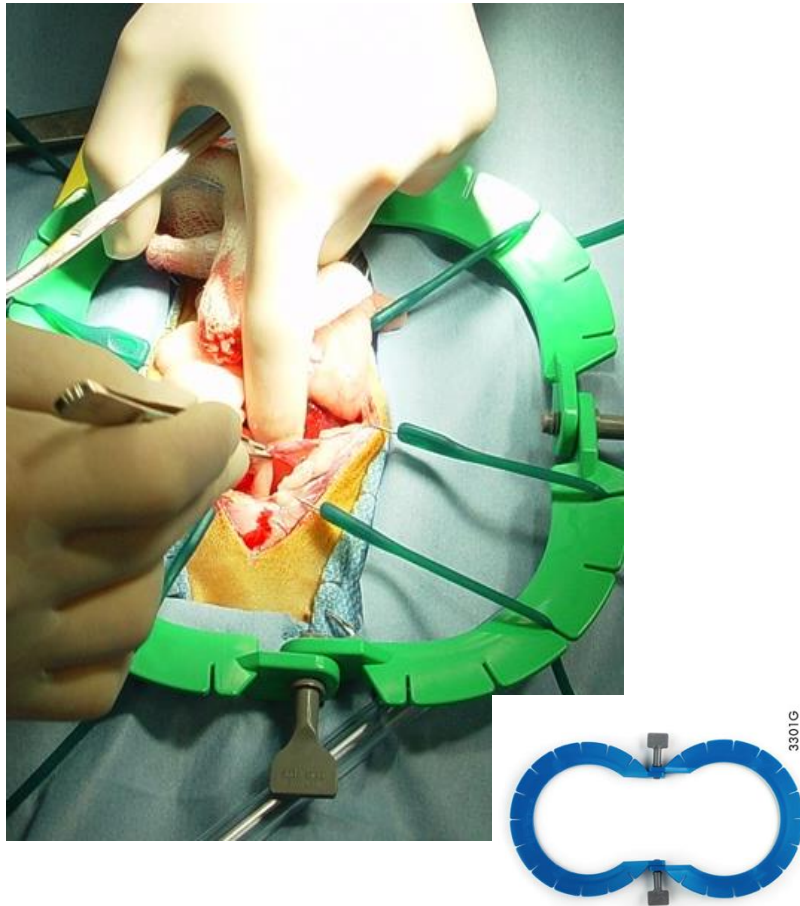


Magnification and Illumination

- Bifocal surgical lens
- Requires practice
- Directional light
- Recommended for surgeries in abdomen



Retractors



- Lone Star retractor
- Autoclavable plastic frame
- Different shapes
- Elastic stays with hooks at the end

Sutures



Table 1. Percentage of retained tensile strength after implantation and number of days required for complete absorption for commonly available suture materials in mammals. Ranges in healing times are shown for different tissues

Suture material	7 Days	14 Days	21 Days	28 Days	42 Days	Complete absorption
Chromic gut			0			60-90 days
Polydioxanone (PDS)		70		50	25	180-210 days
Polyglyconate (Maxon)	80	75	65	50	25	180 days
Glycomer 631 (Biosyn)		75	40			90-110 days
Polyglecaprone 25 (Monocryl)	60-70	30-40				90-120 days
Polyglycolic acid (Dexon)		20				60 days
Polyglactin 910 (Vicryl)		75	50	25		56-70 days
Lactomer (Polysorb)		80	30			56-70 days
Polyglytone 6211 (Caprosyn)	50-60	20-30	0			56 days
Polyglactin 910 (Vicryl Rapide)	50	0				42 days
Bladder		X-----X				
Gastrointestinal tract		X-----X (SI)				
		X-----X (LI)				
Fascia		X-----X				

Abbreviations: SI, Small intestine; LI, large intestine.

Vascular Clips



- Steel, titanium
- Quick and easy
- Various sizes
- Straight and angle

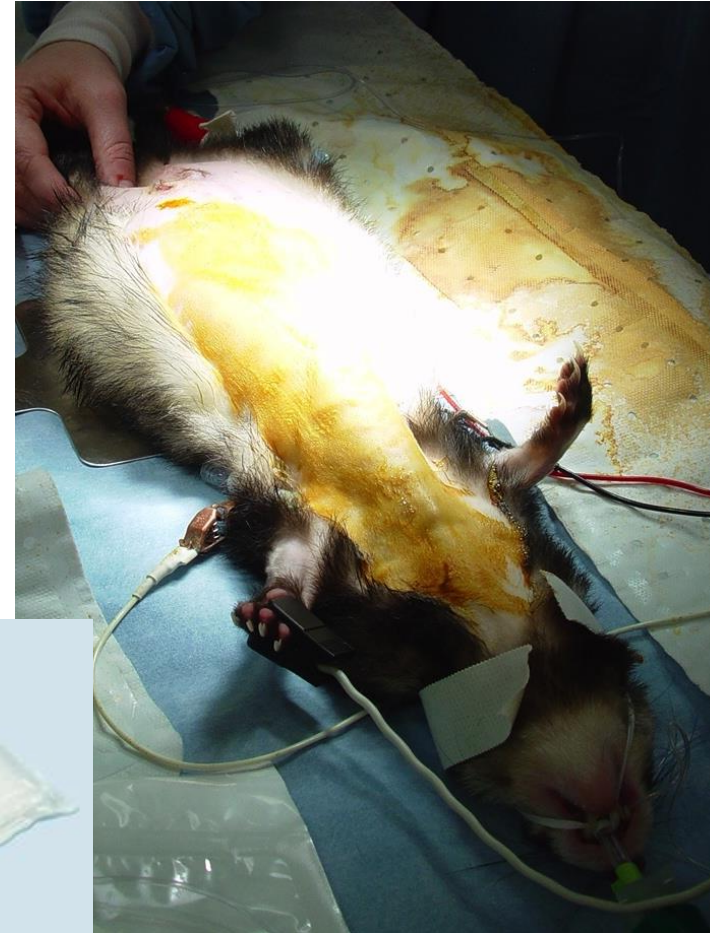
Tissue Sealers

- Ligasure® open sealer
- Grasp, seals and cut



Patient Preparation

- Laborious clipping
- Chlorhexidine and saline
- Clear drapes
- Quarter and larger drape



Rabbit

- Facial abscess
- Partial ear canal ablation lat. bulla osteotomy
- Liver lobectomy
- Coxofemoral amputation
- Ovariohysterectomy
- Orchidectomy

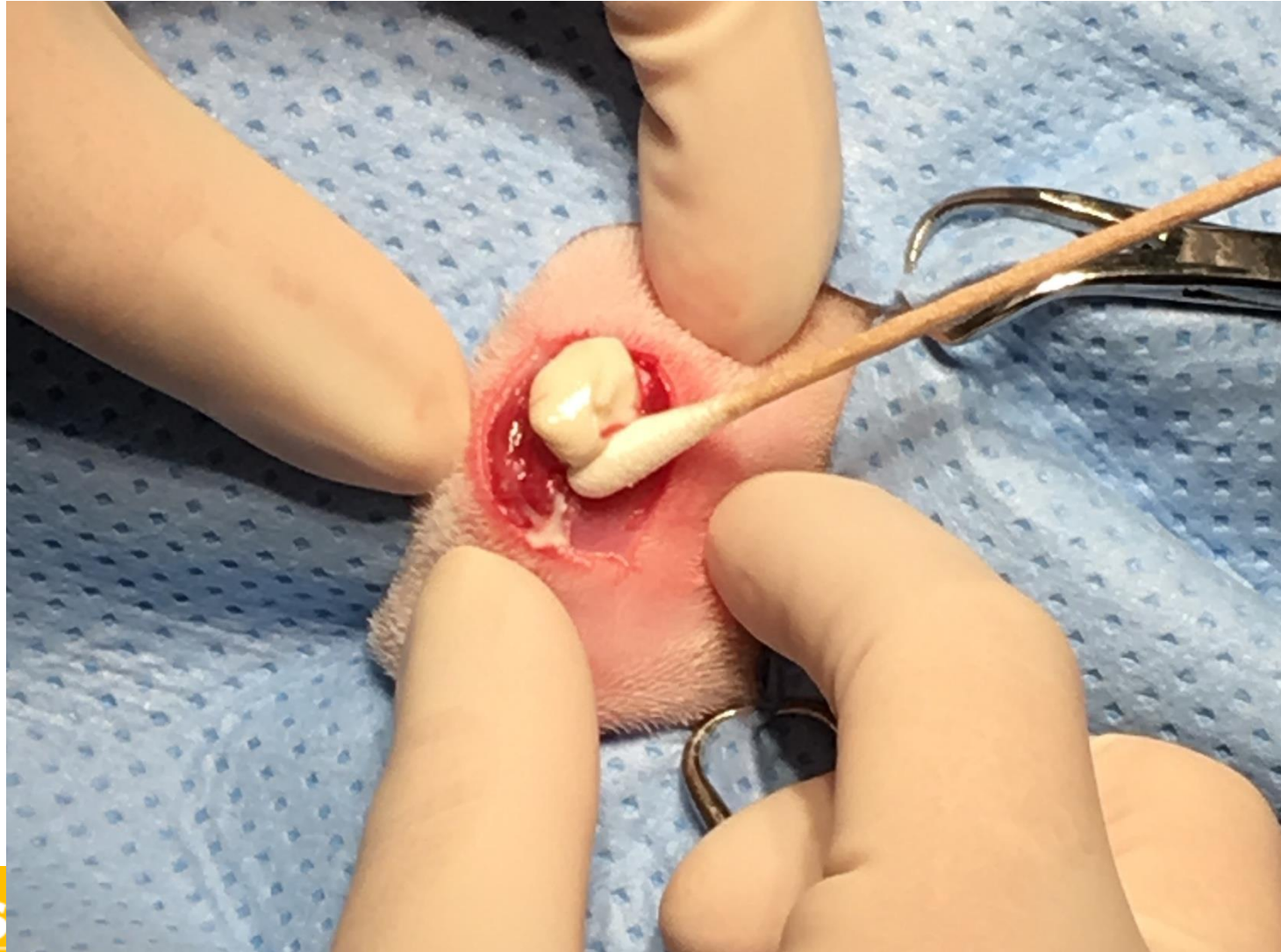
Rabbit- Facial Abscess

- Patient on dorsal or lateral recumbency
- Skin incision over the abscess and extend incision to abscess capsule
- Remove purulent exudate, sample capsule for culture and susceptibility and flush copiously
- Debride osteomyelitic bone if needed
- Extend edges incision and suture capsule to skin with simple interrupted

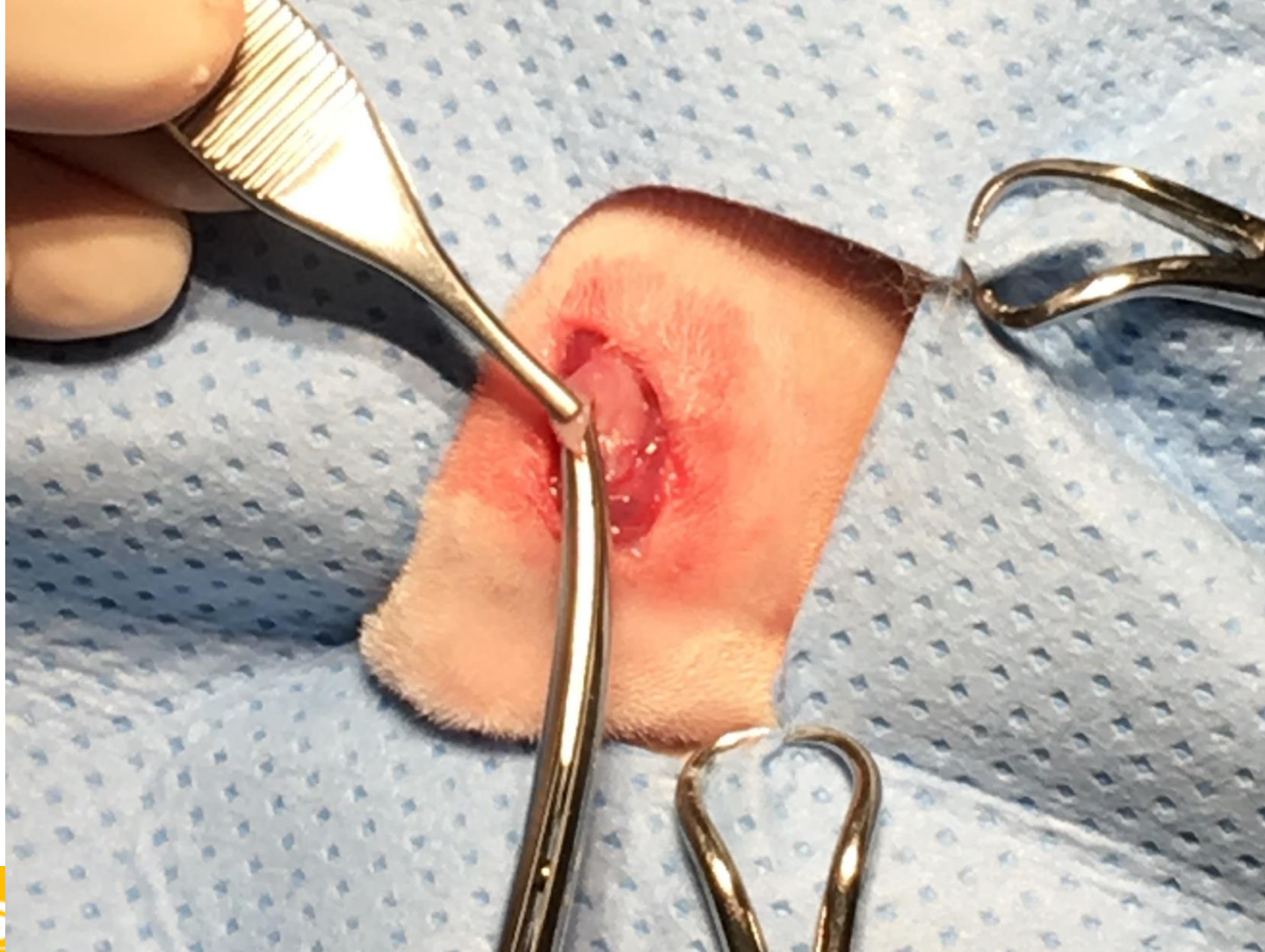
Rabbit- Facial Abscess



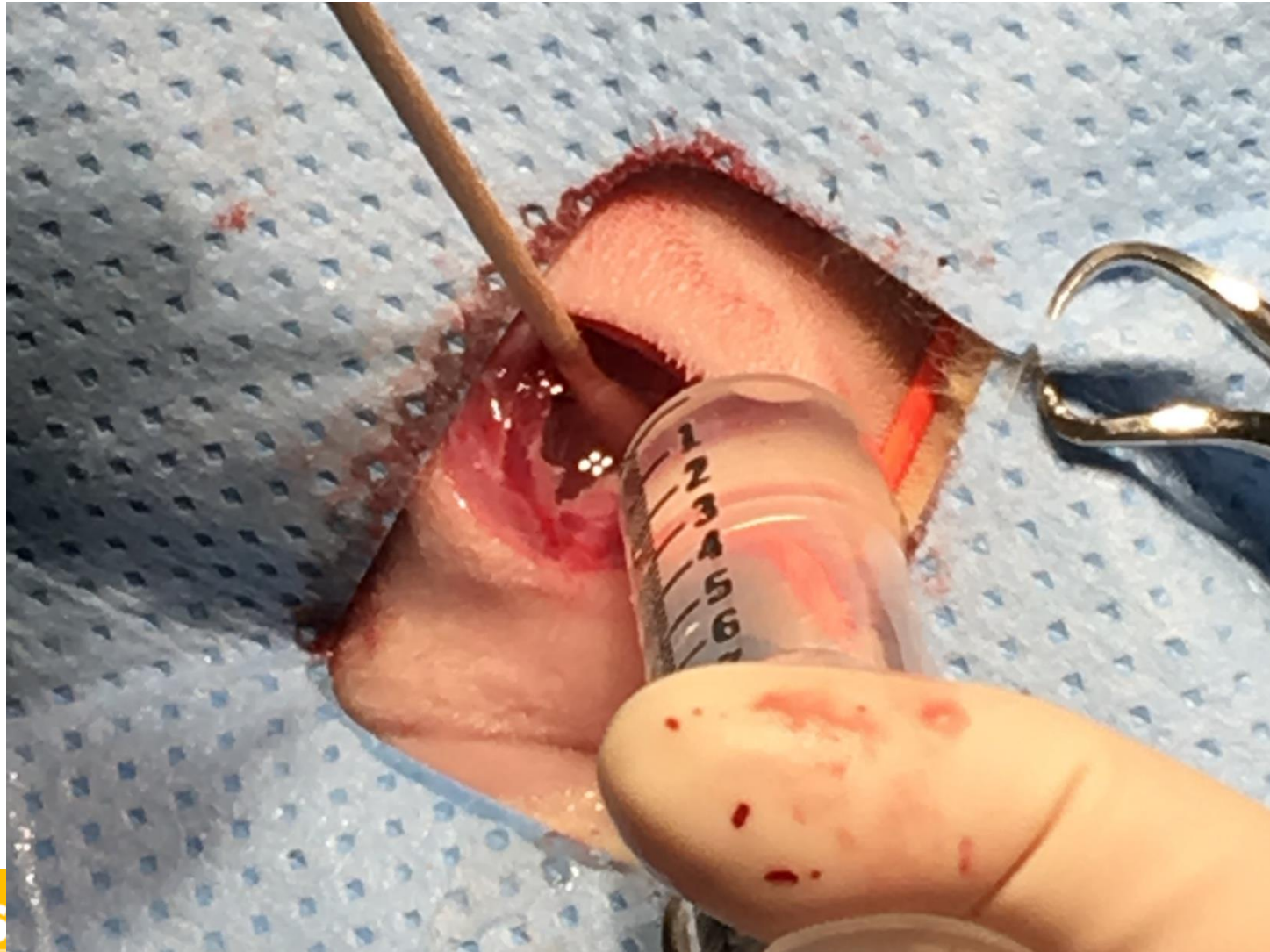
Rabbit- Facial Abscess



Rabbit- Facial Abscess



Rabbit- Facial Abscess



Rabbit- Facial Abscess



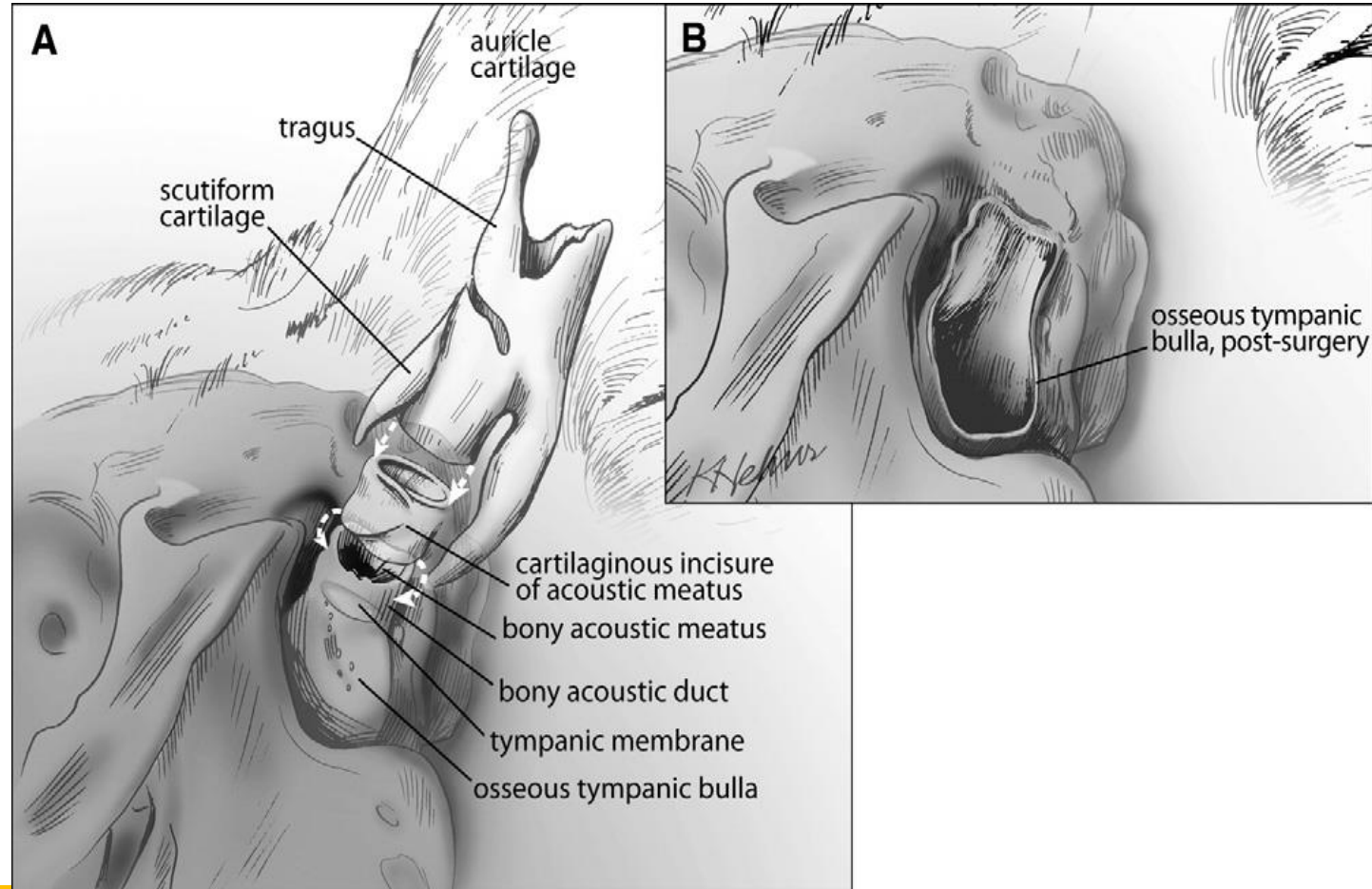
Rabbit- Facial Abscess



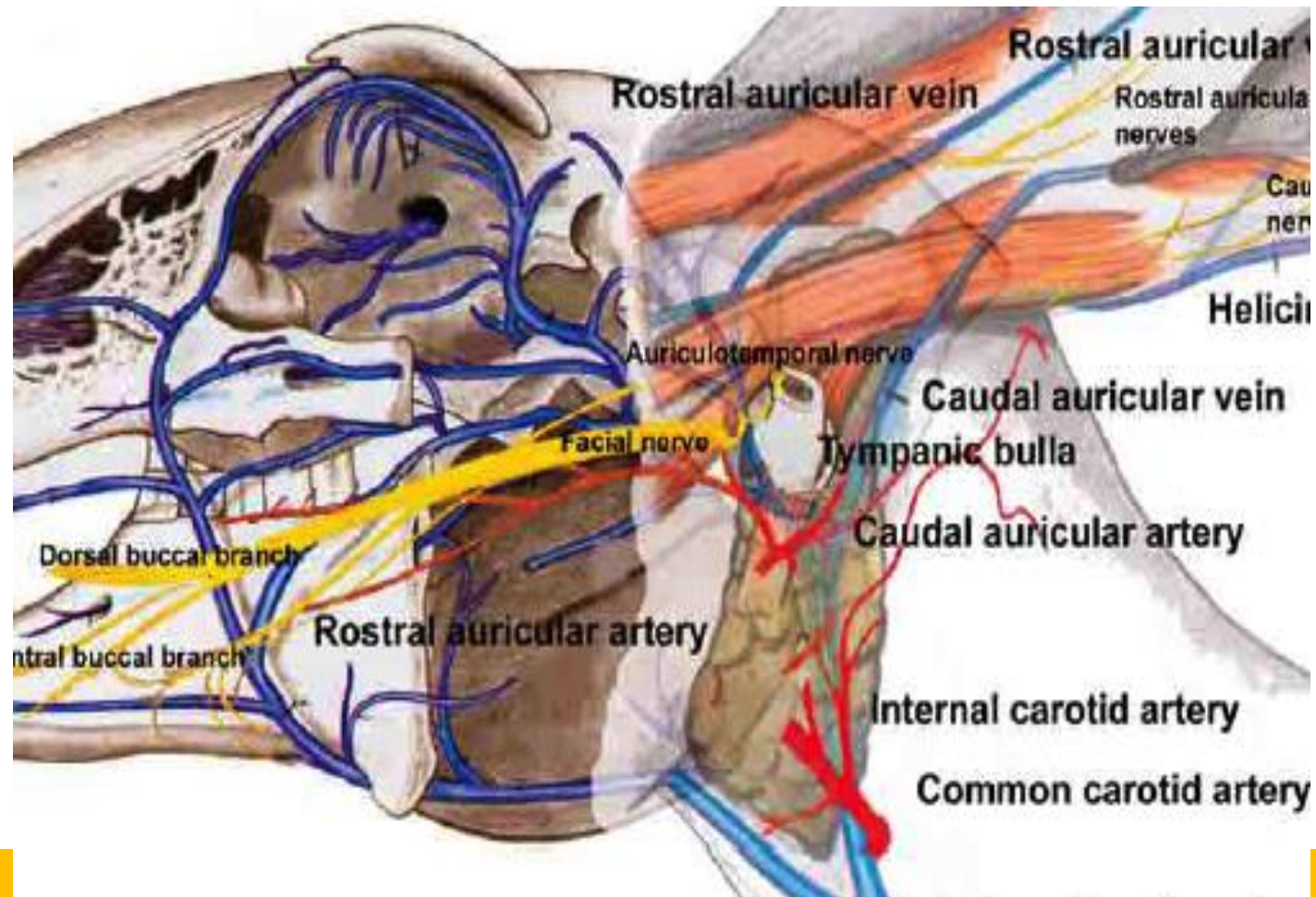
Rabbit- Partial Ear Canal Ablation Lateral Bulla Osteotomy

- In lateral recumbency, incise skin over ear canal to cartilage or mucosa, dissect around canal, transect distal canal
- Dissect ear canal to the bony acoustic meatus, transect ear canal, collect samples for C&S
- Dissect soft tissues over lateral wall of bulla with care, remove bone with rongeurs, clean bulla with care and flush
- Fill bulla with poloxamer gel with antibiotics
- Close subcutaneous tissue and skin

Rabbit- Partial Ear Canal Ablation Lateral Bulla Osteotomy



Rabbit- Partial Ear Canal Ablation Lateral Bulla Osteotomy



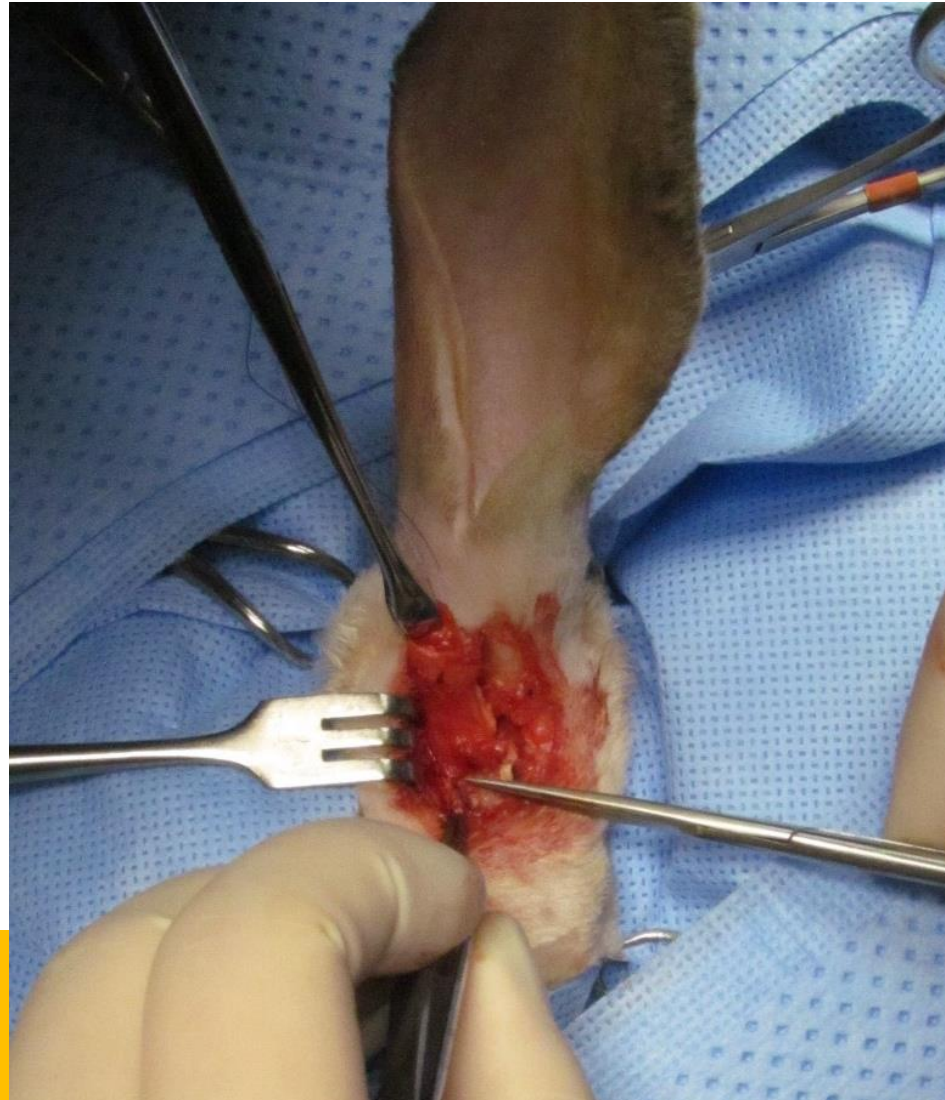
Rabbit- Partial Ear Canal Ablation Lateral Bulla Osteotomy



Rabbit- Partial Ear Canal Ablation Lateral Bulla Osteotomy



Rabbit- Partial Ear Canal Ablation Lateral Bulla Osteotomy



Rabbit- Partial Ear Canal Ablation Lateral Bulla Osteotomy



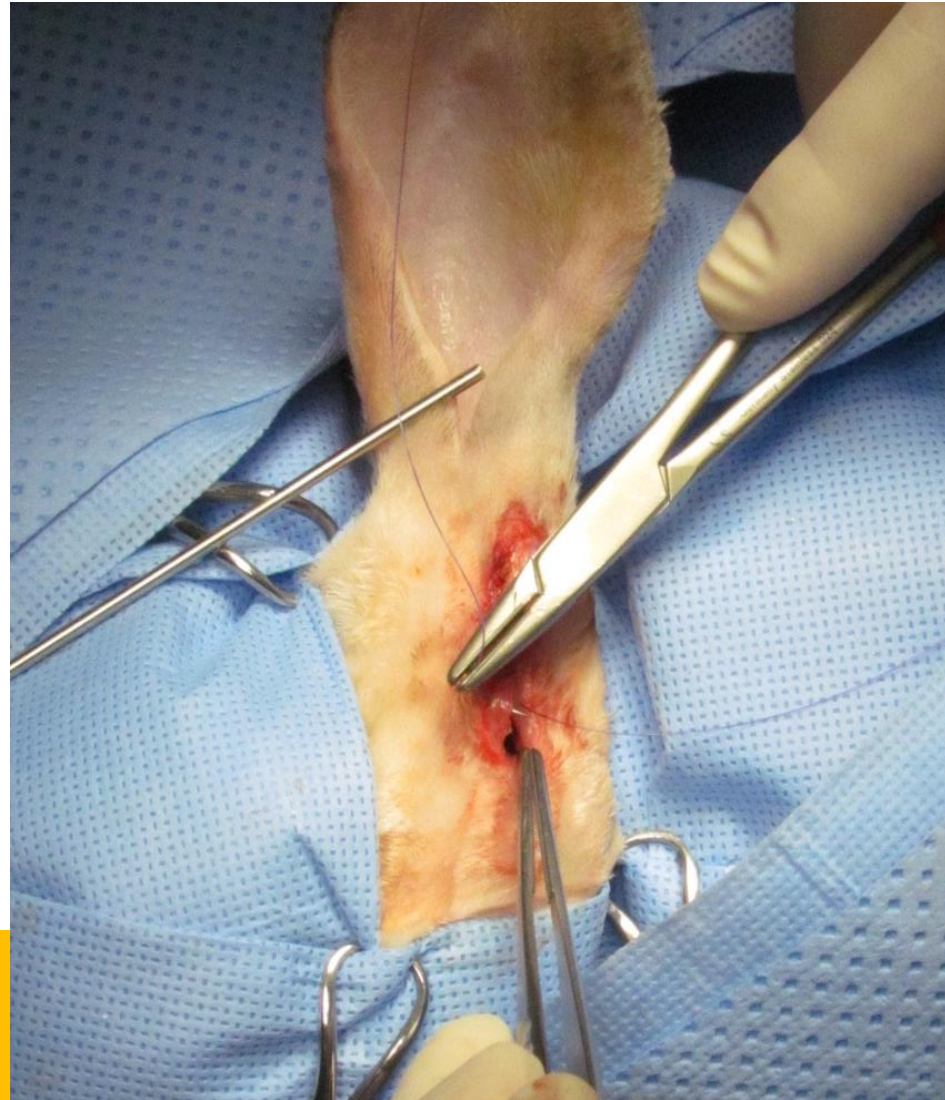
Rabbit- Partial Ear Canal Ablation Lateral Bulla Osteotomy



Rabbit- Partial Ear Canal Ablation Lateral Bulla Osteotomy



Rabbit- Partial Ear Canal Ablation Lateral Bulla Osteotomy



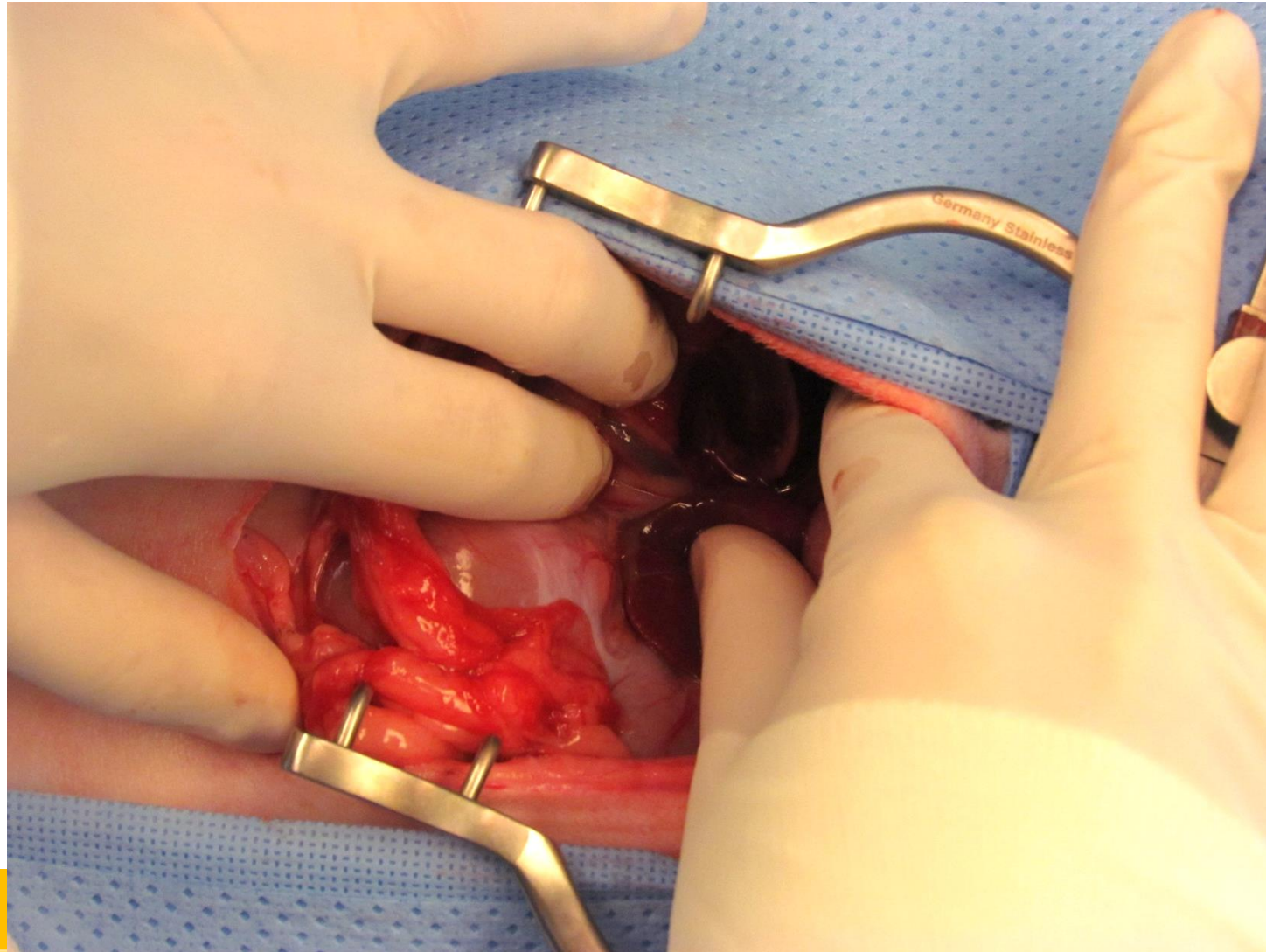
Rabbit- Partial Ear Canal Ablation Lateral Bulla Osteotomy



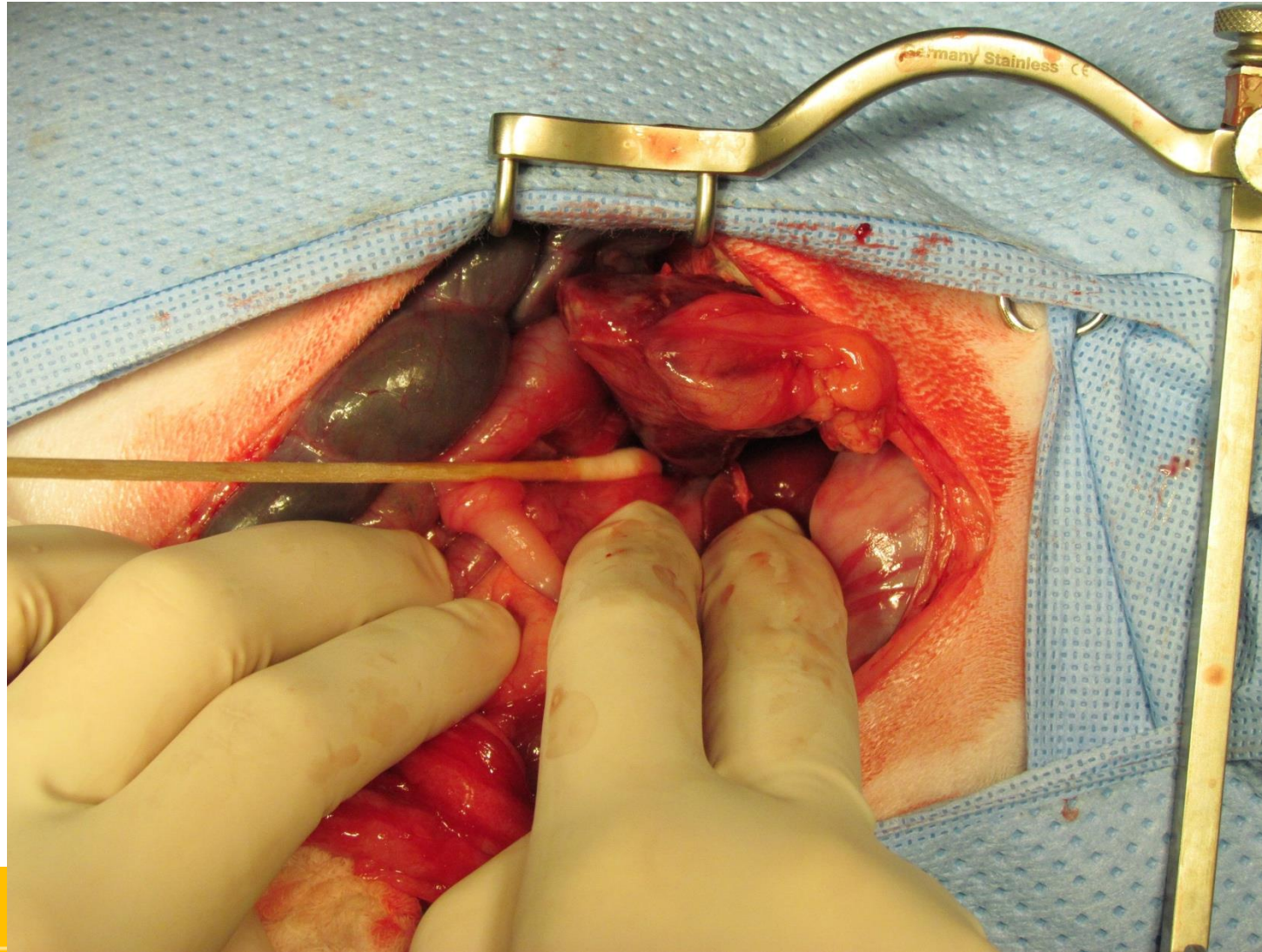
Rabbit- Liver Lobectomy

- Skin incision midline between umbilicus and xyphoid, then linea alba incision
- Identify liver lobe affected
- Ligate x2 vessels and bile ducts at the base of the lobe, transected distally
- Close fascia, SC tissue and skin (subcuticular)

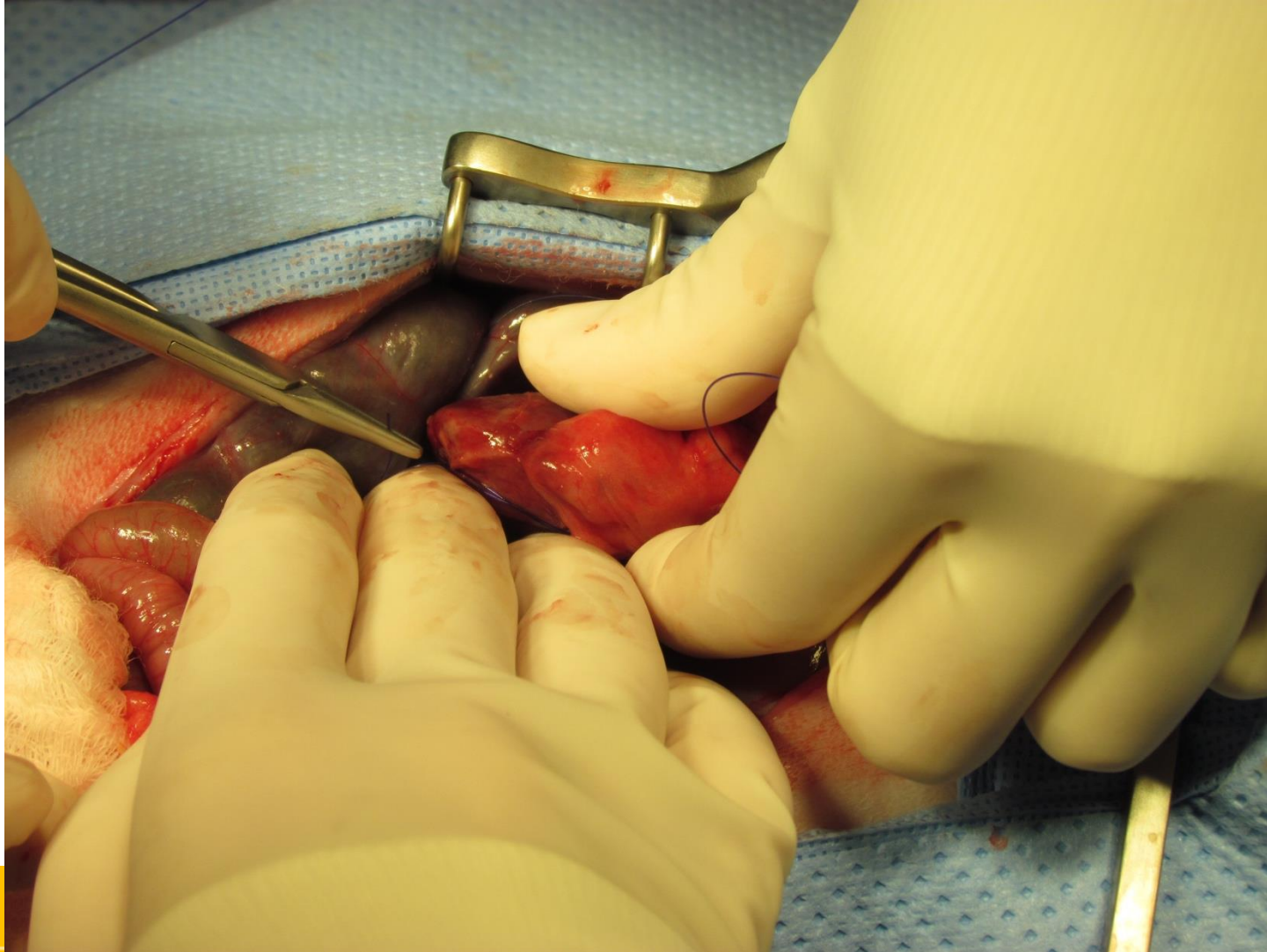
Rabbit- Lobectomy



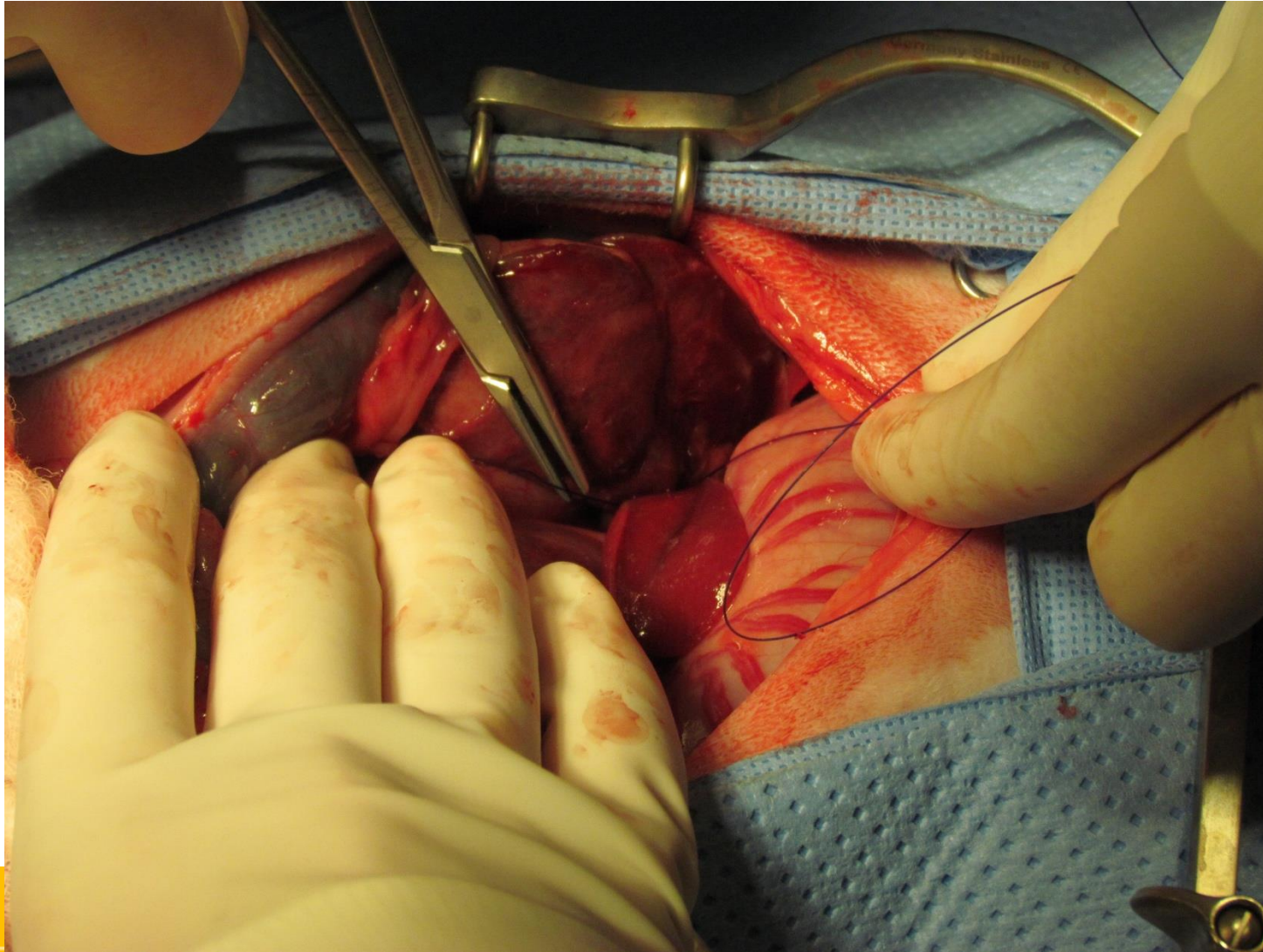
Rabbit- Lobectomy



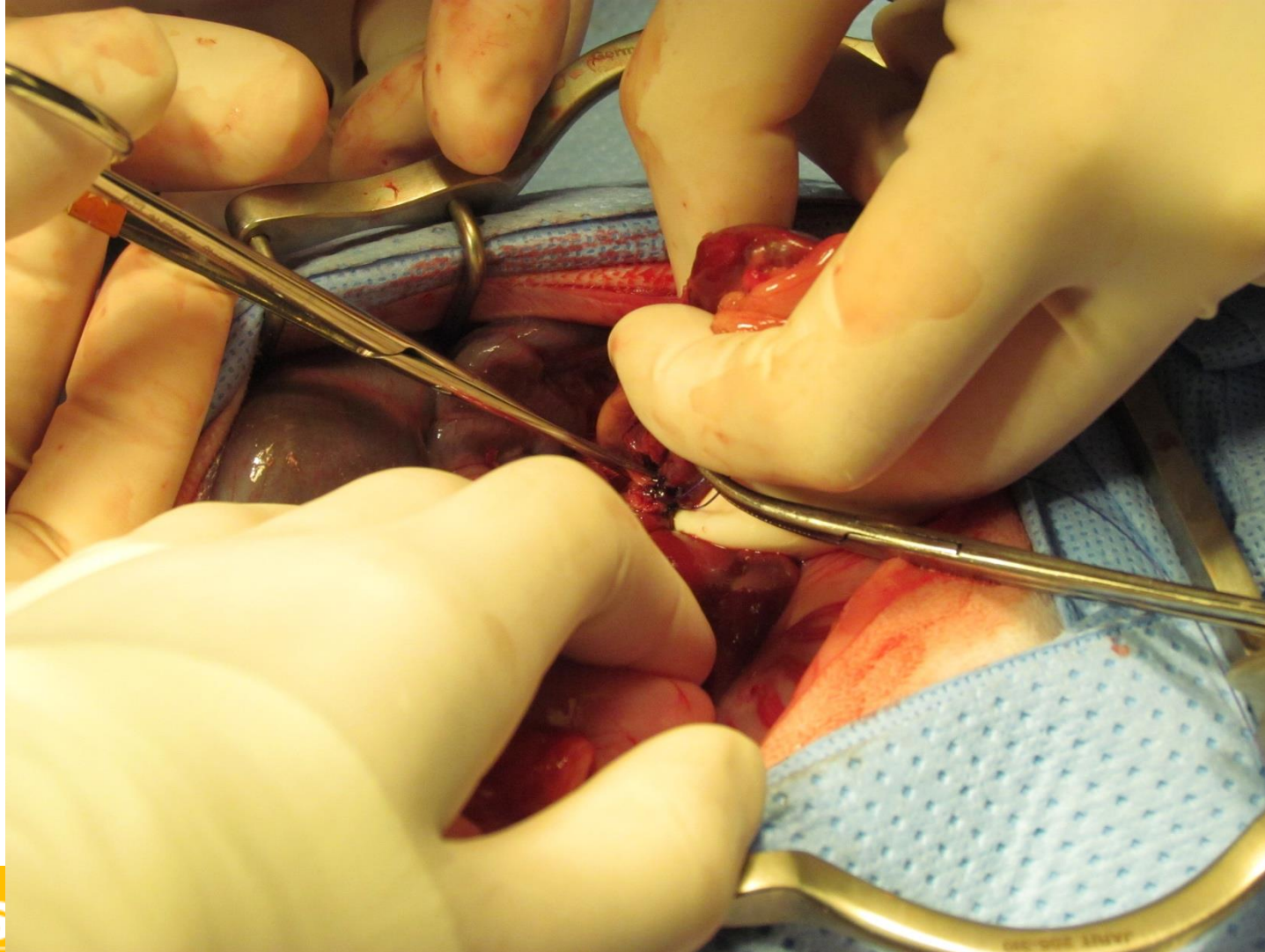
Rabbit- Lobectomy



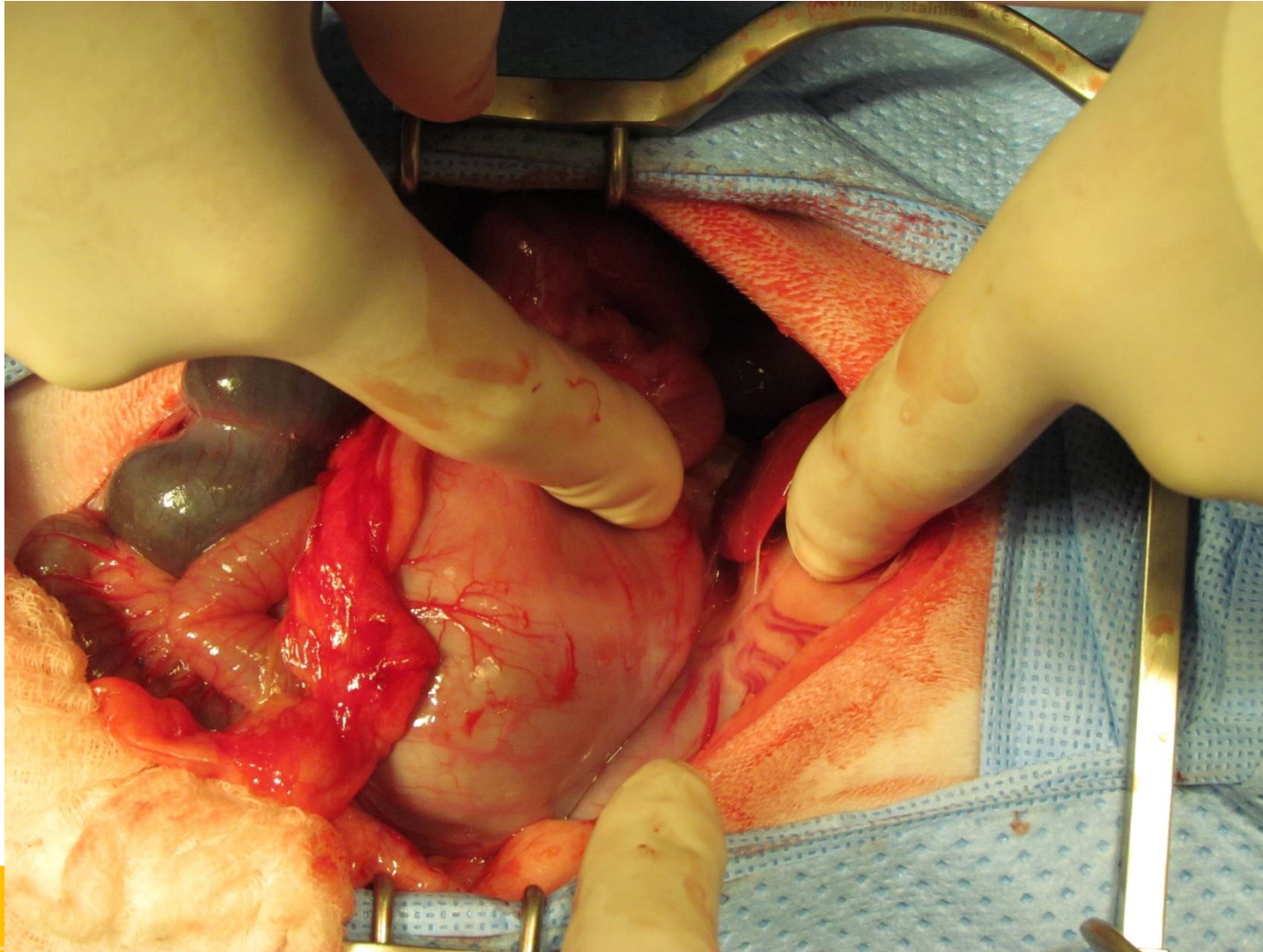
Rabbit- Lobectomy



Rabbit- Lobectomy



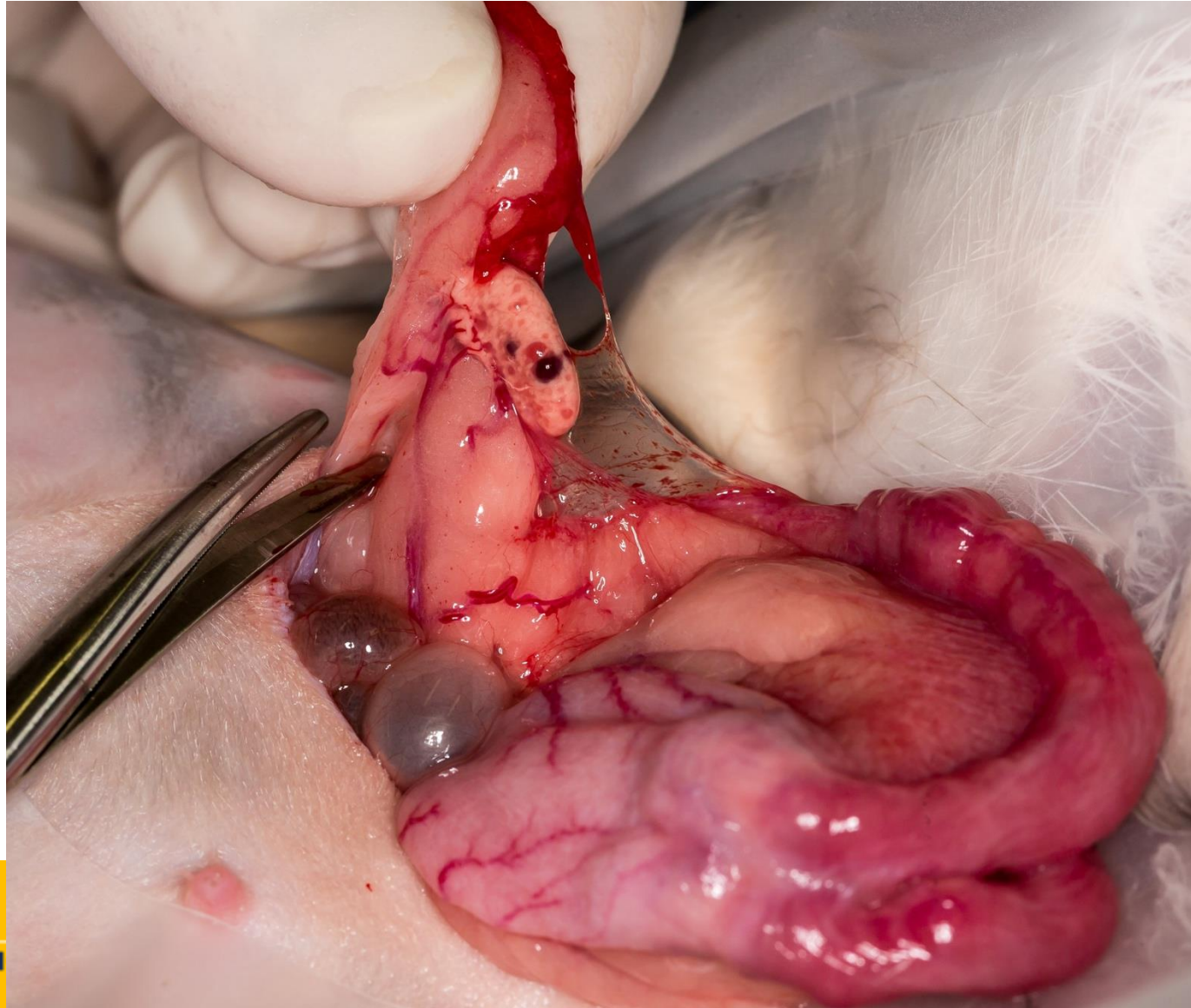
Rabbit- Lobectomy



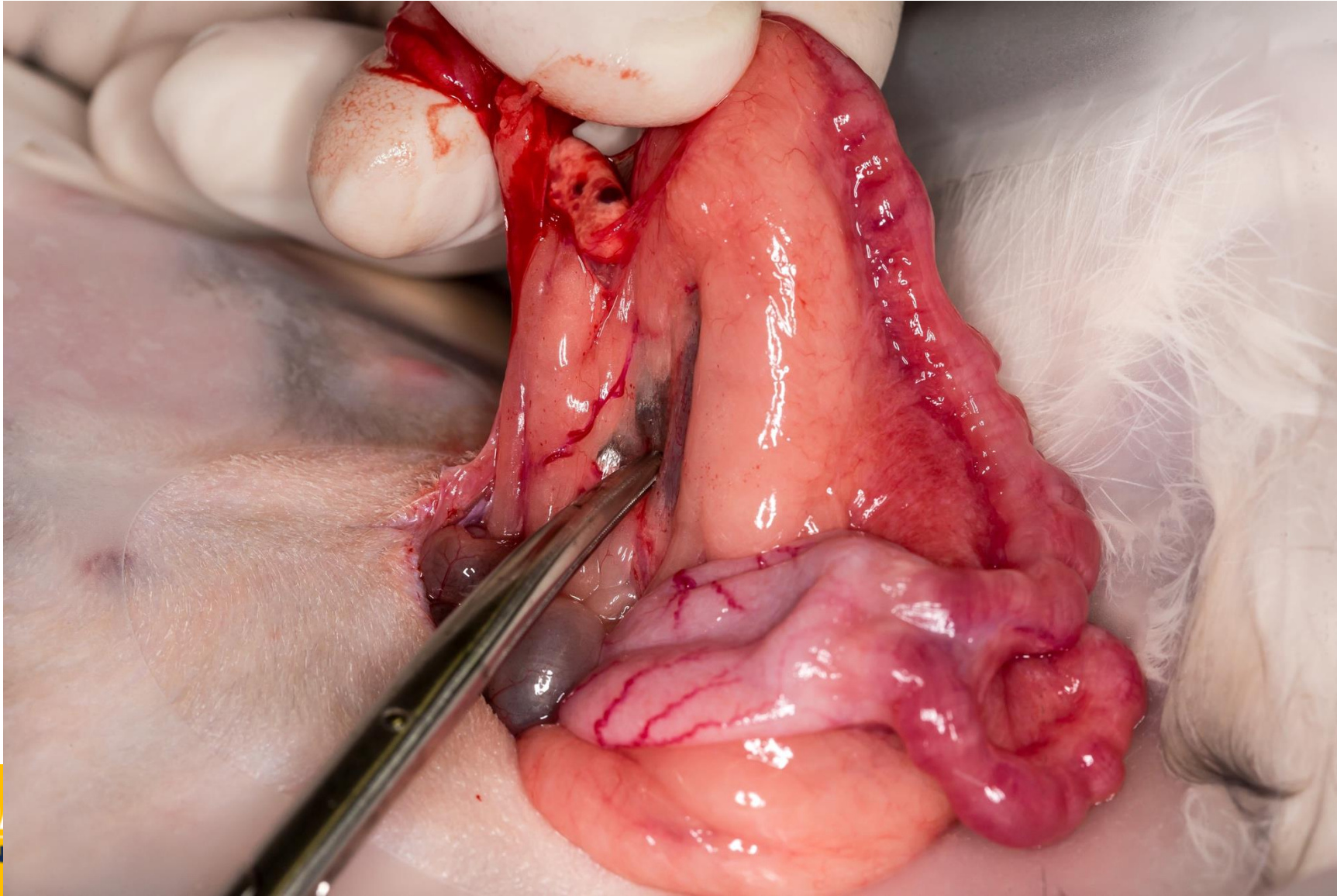
Rabbit- Ovariohysterectomy

- Skin incision midline between umbilicus and pubis (3-4 cm), then linea alba incision
- Exteriorize uterine horns, identify ovaries
- Ligate x2 L ovarian pedicle and transect, same on L mesometrium; repeat on R side
- Ligate x2 caudal cervixes or cranial vagina
- Close fascia, SC tissue and skin (subcuticular)

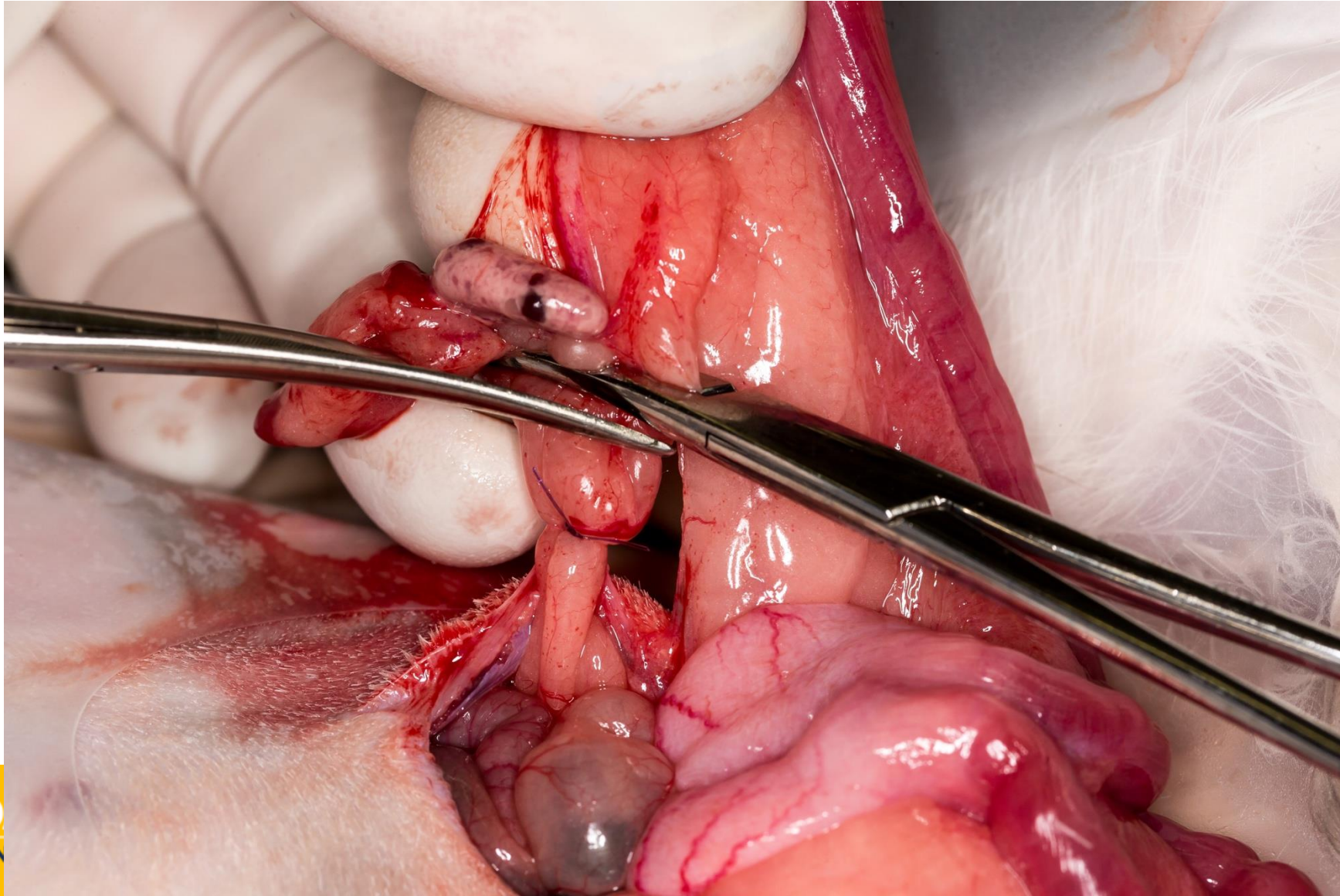
Rabbit- Ovariohysterectomy



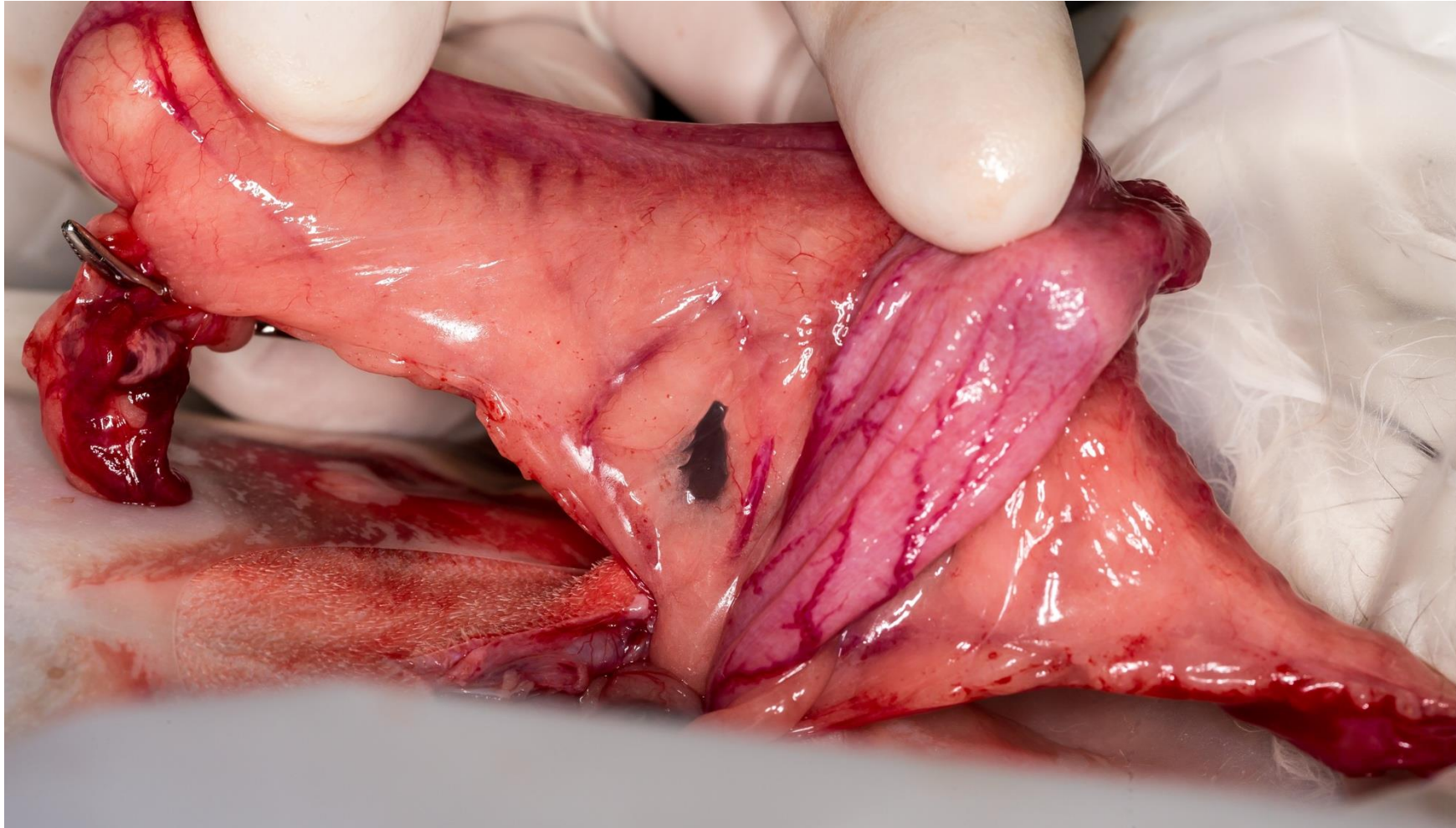
Rabbit- Ovariohysterectomy



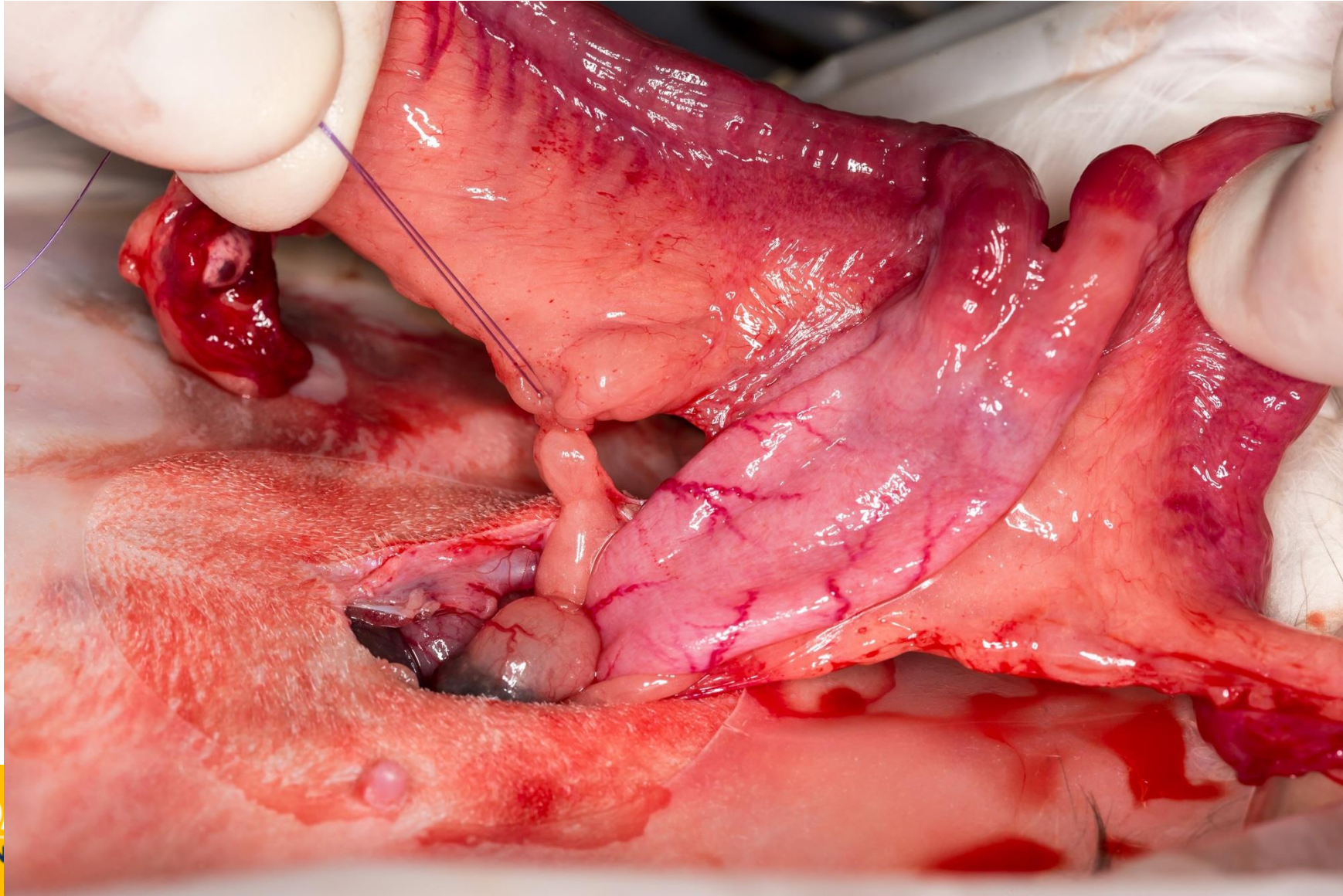
Rabbit- Ovariohysterectomy



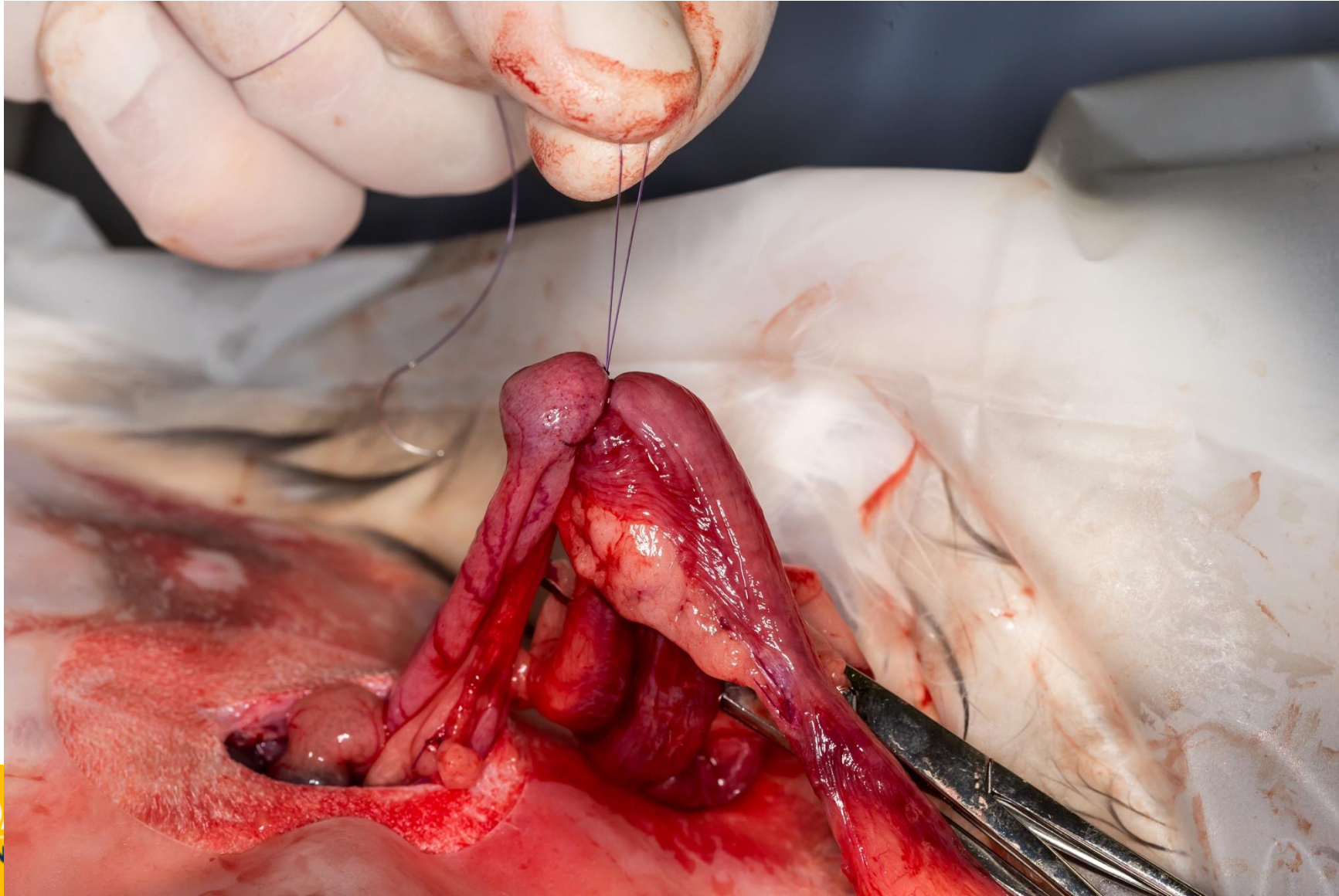
Rabbit- Ovariohysterectomy



Rabbit- Ovariohysterectomy



Rabbit- Ovariohysterectomy



Rabbit- Orchidectomy

Scrotal Open Technique

- Skin incision over scrotum from Cr to Ca incising the vaginal tunic
- Exteriorize and retract the testicle placing hemostat across vaginal tunic
- Separate, clamp and ligate the vascular cord and ductus deferens X 2 and transect them
- Ligate vaginal tunic
- Skin closure with subcuticular or tissue glue

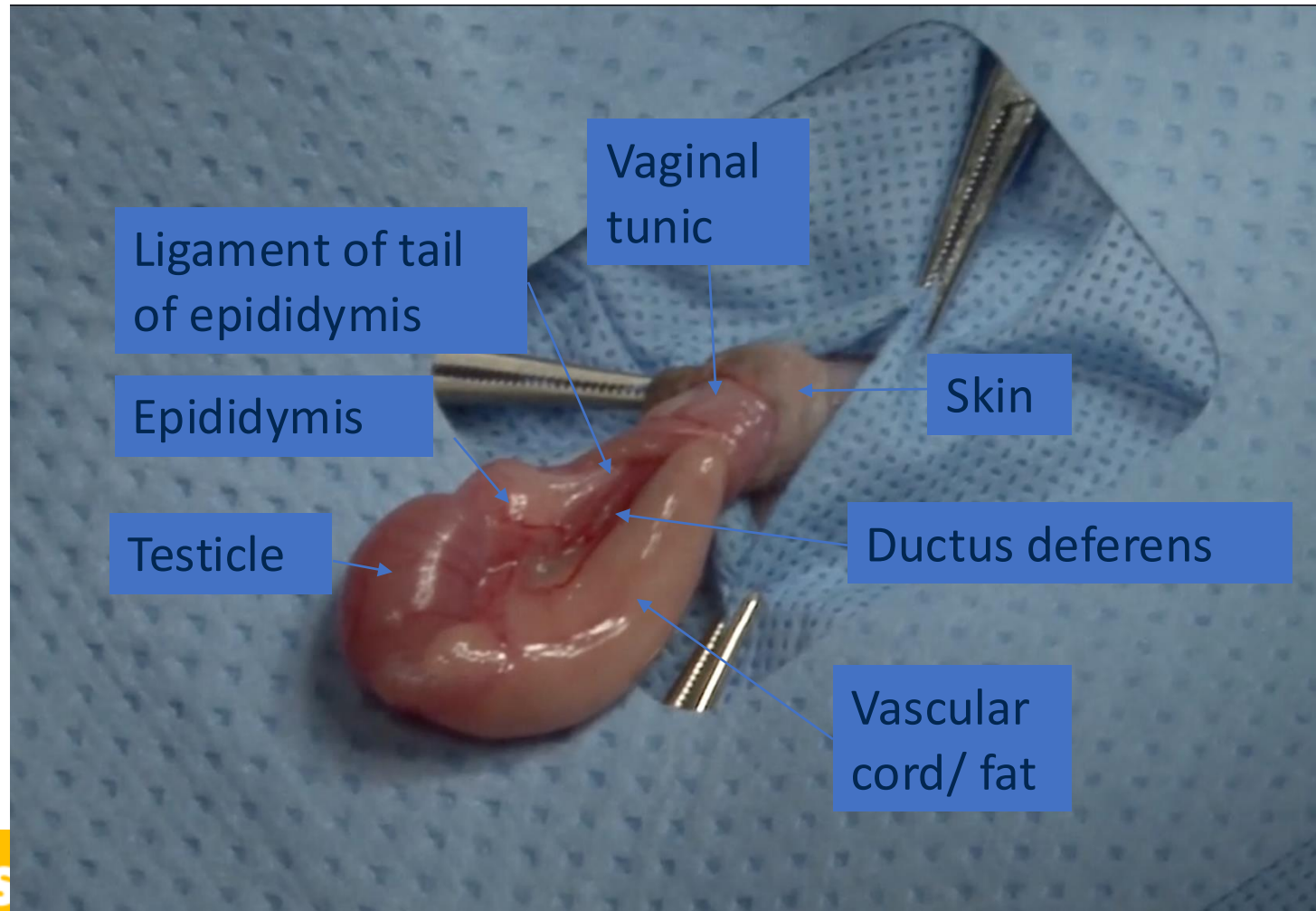
Rabbit- Orchidectomy

Scrotal Open Technique



Rabbit- Orchidectomy

Scrotal Open Technique



Rabbit- Orchidectomy

Scrotal Open Technique



Rabbit- Orchidectomy

Scrotal Open Technique



Rabbit- Orchidectomy

Scrotal Open Technique



Rabbit- Orchidectomy

Scrotal Open Technique



Rabbit- Orchidectomy

Scrotal Open Technique



Ferret

- Gastrointestinal surgery
- Adrenalectomy
- Pancreatectomy

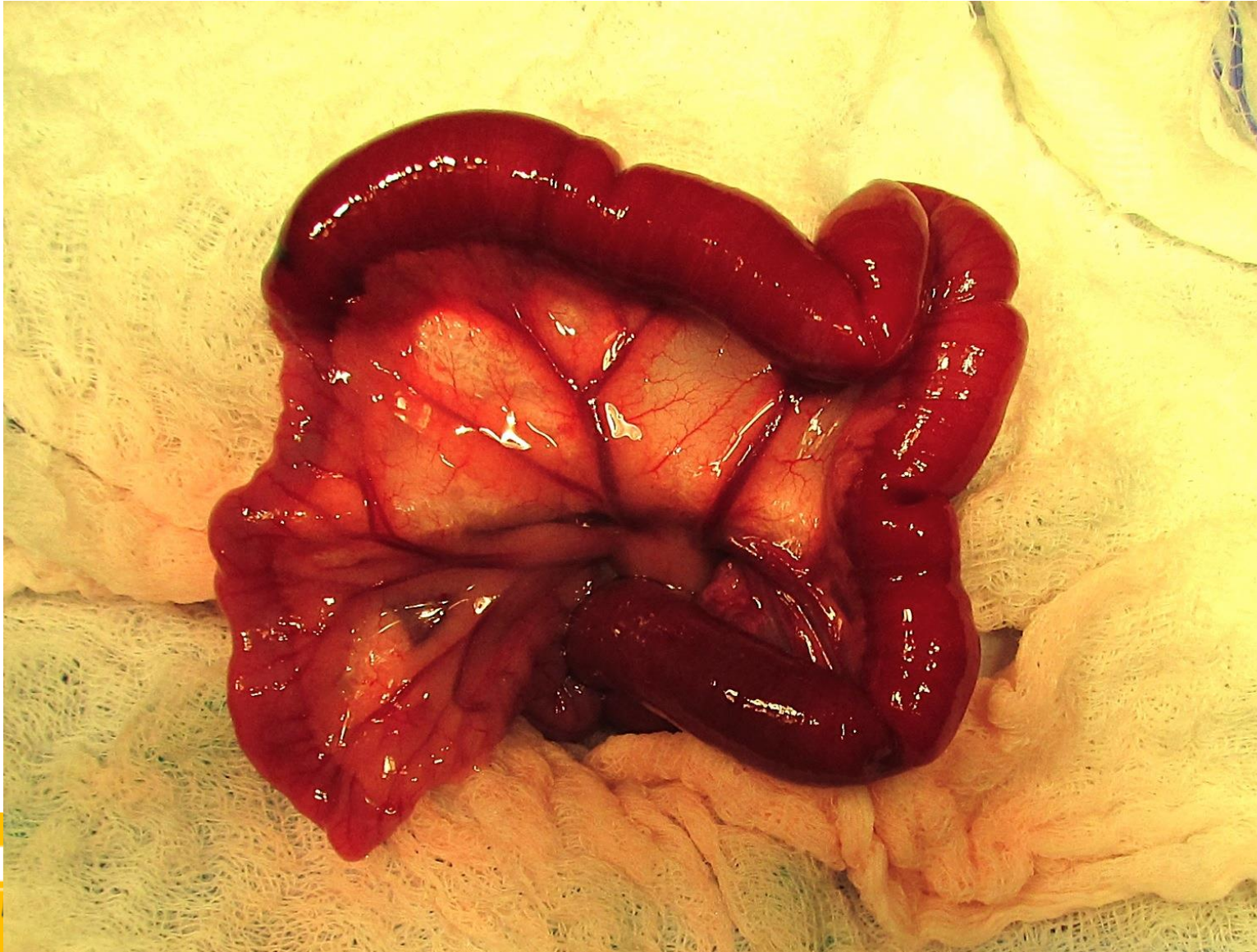
Ferret- Gastrointestinal Surgery



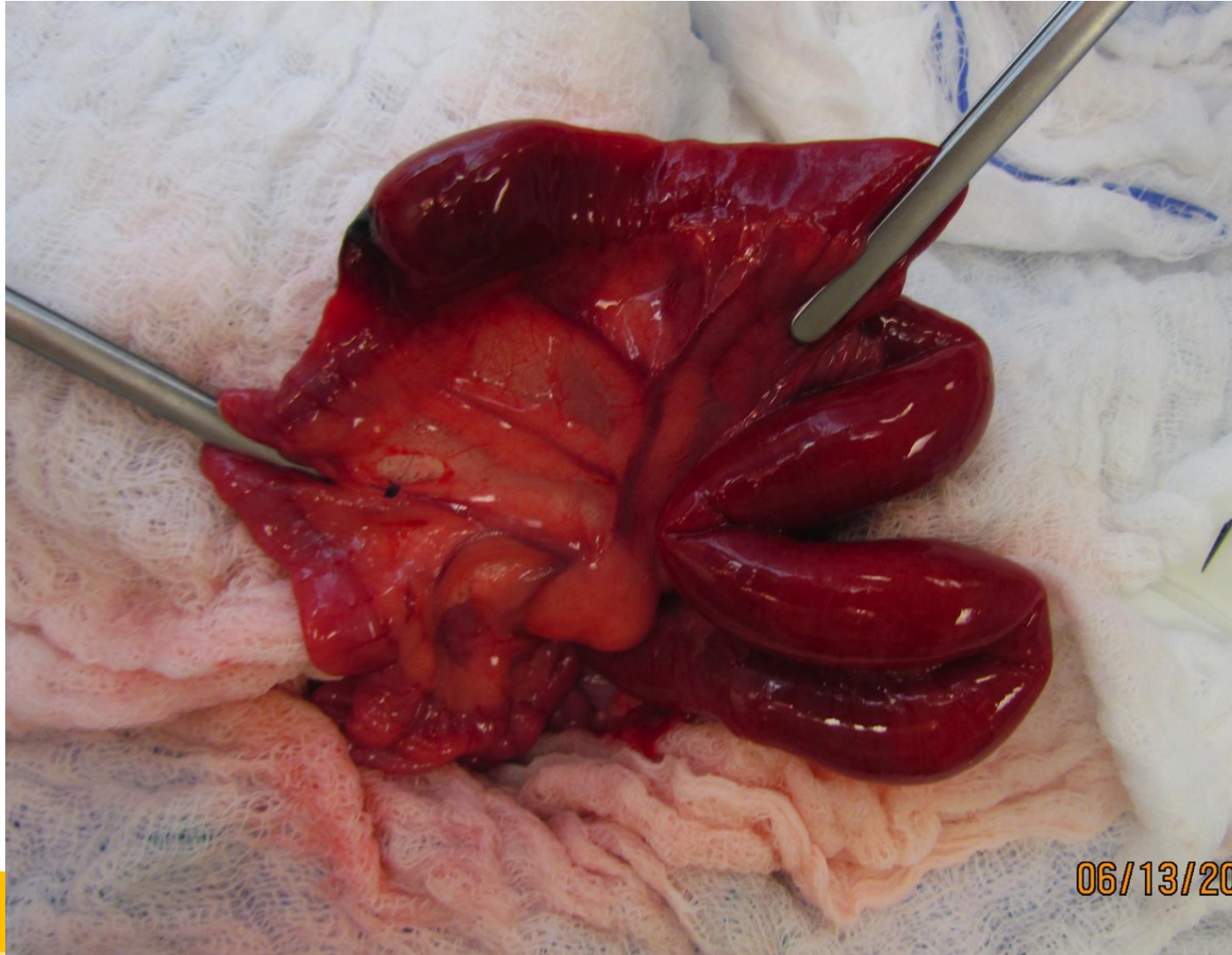
Ferret- Gastrointestinal Surgery



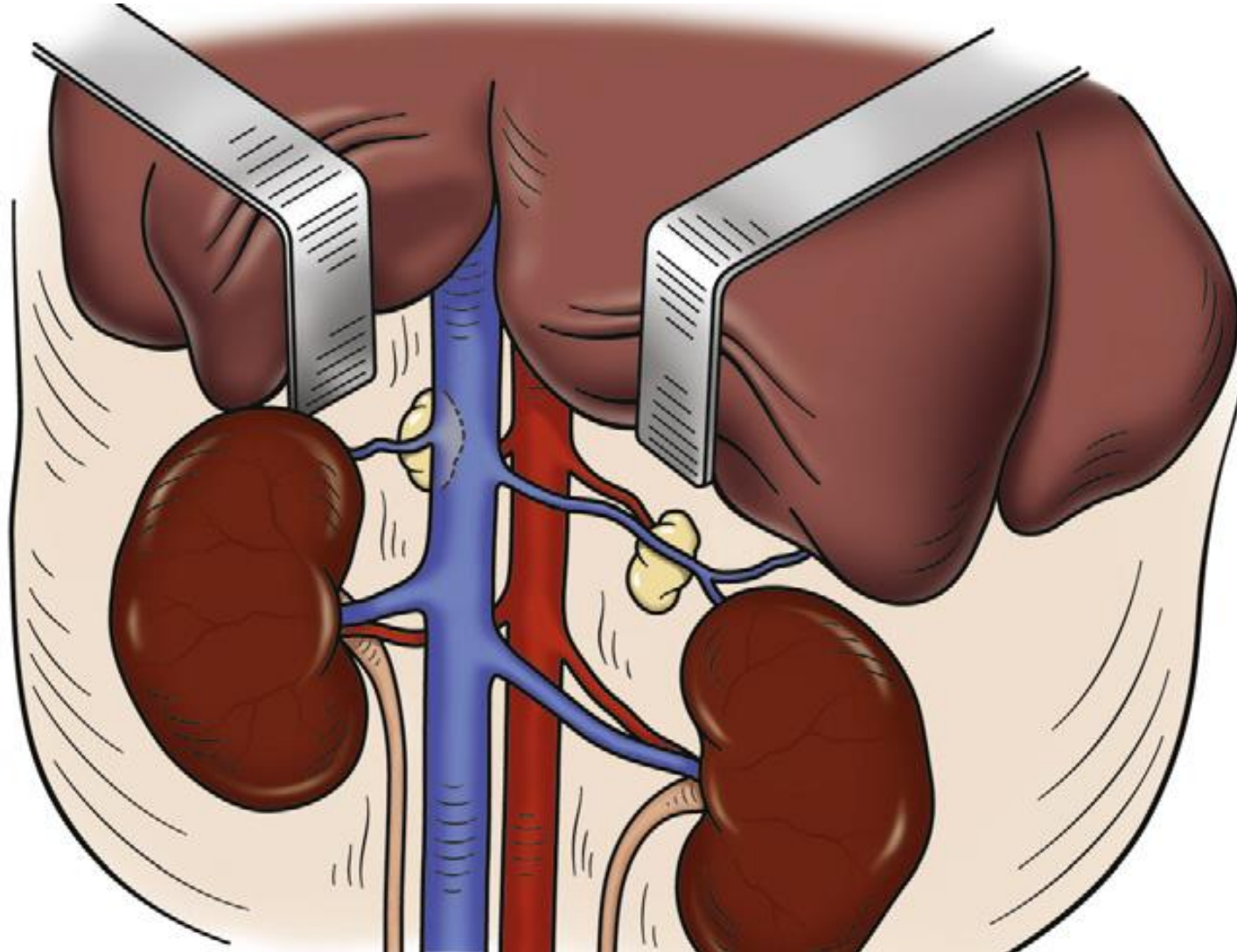
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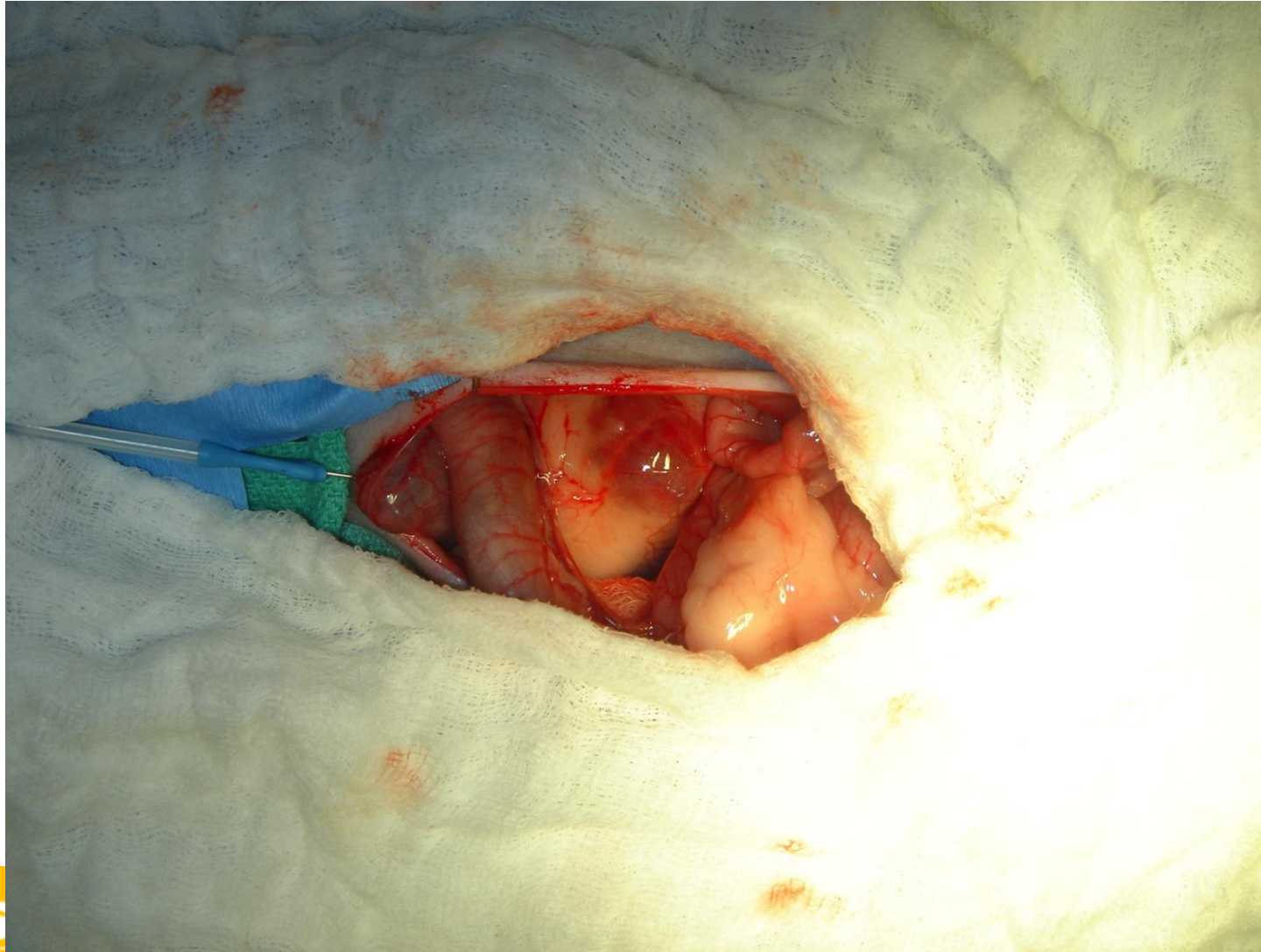
Ferret- Gastrointestinal Surgery



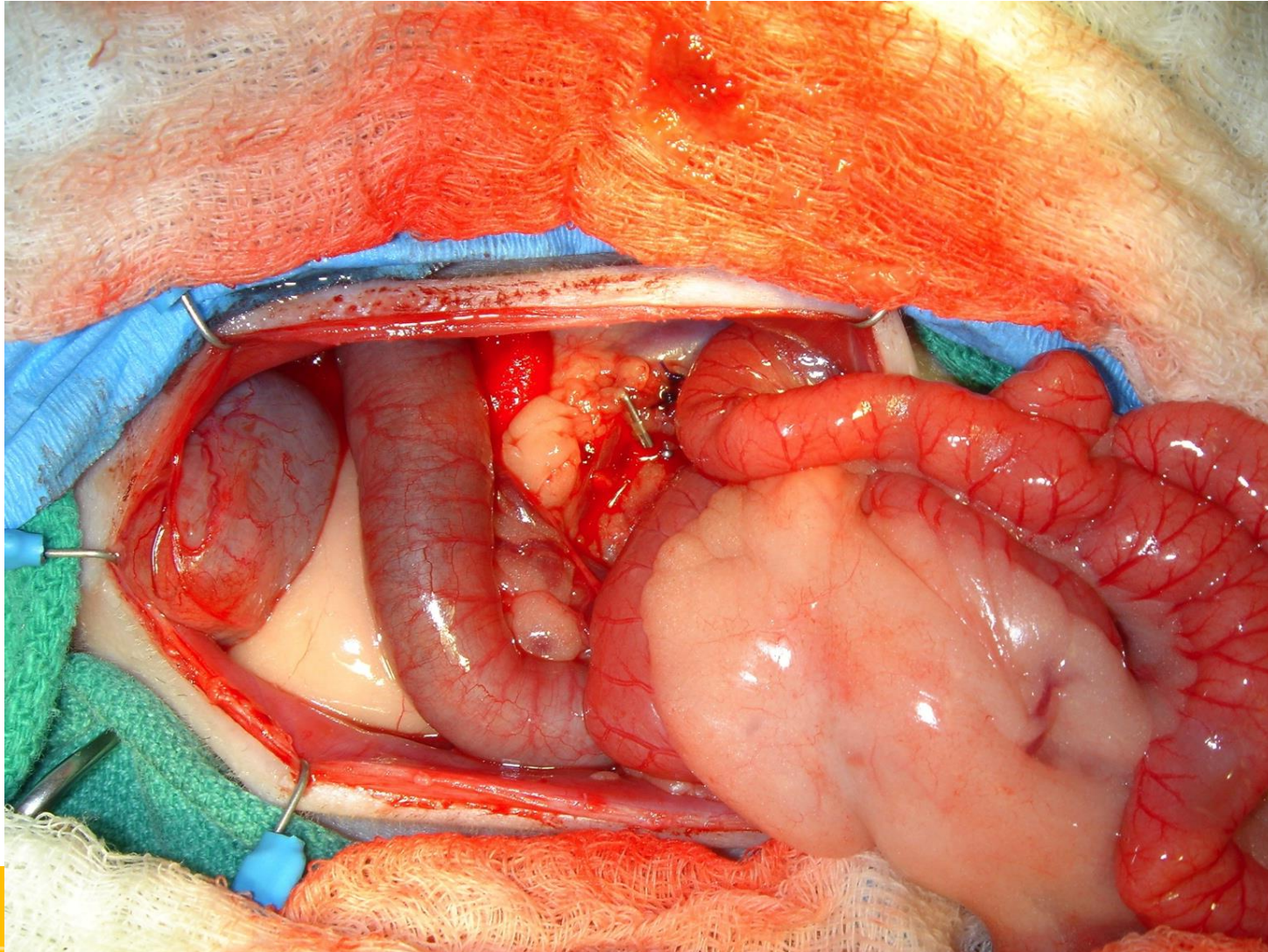
Ferret- Adrenalectomy



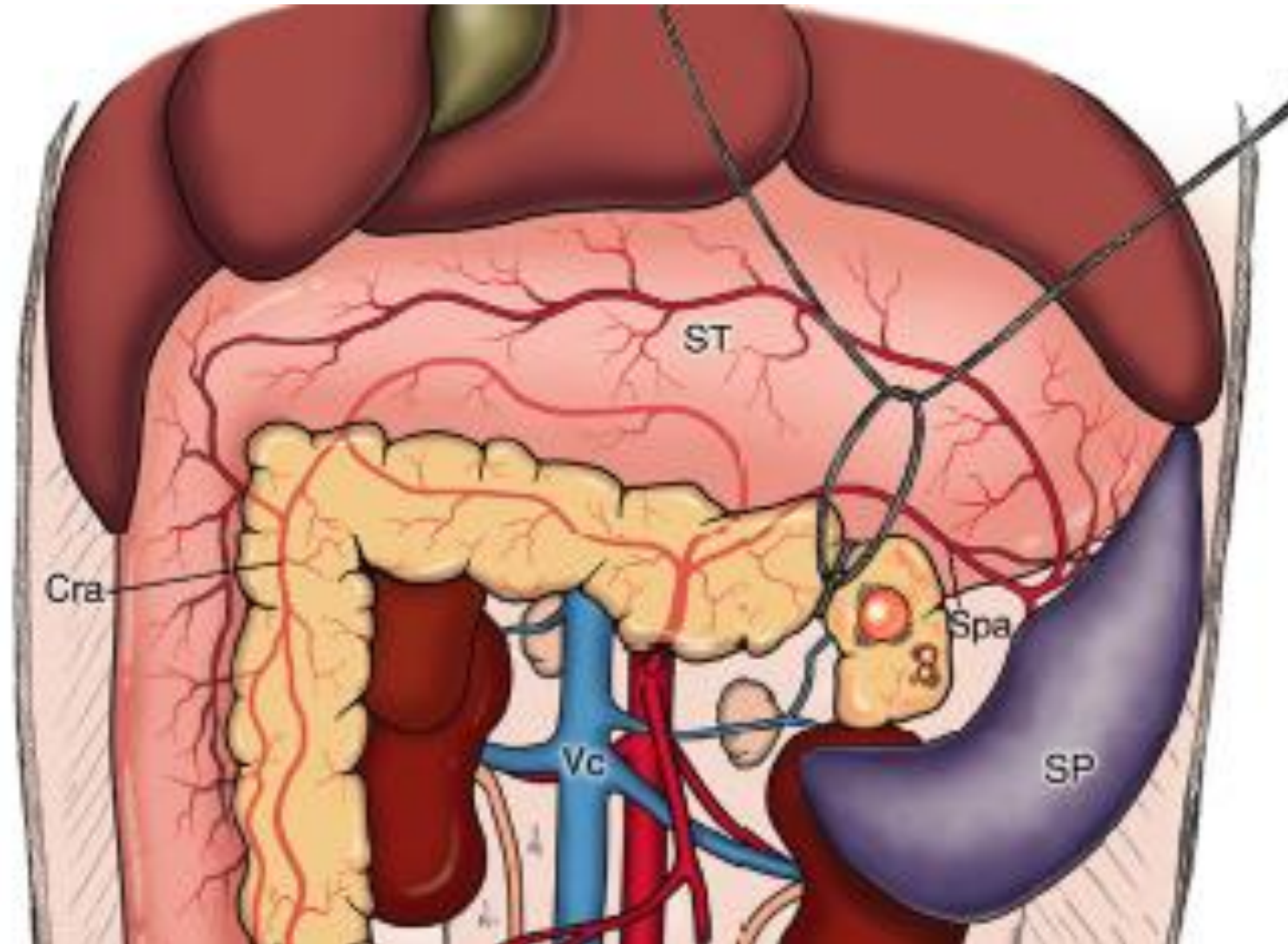
Ferret- Adrenalectomy



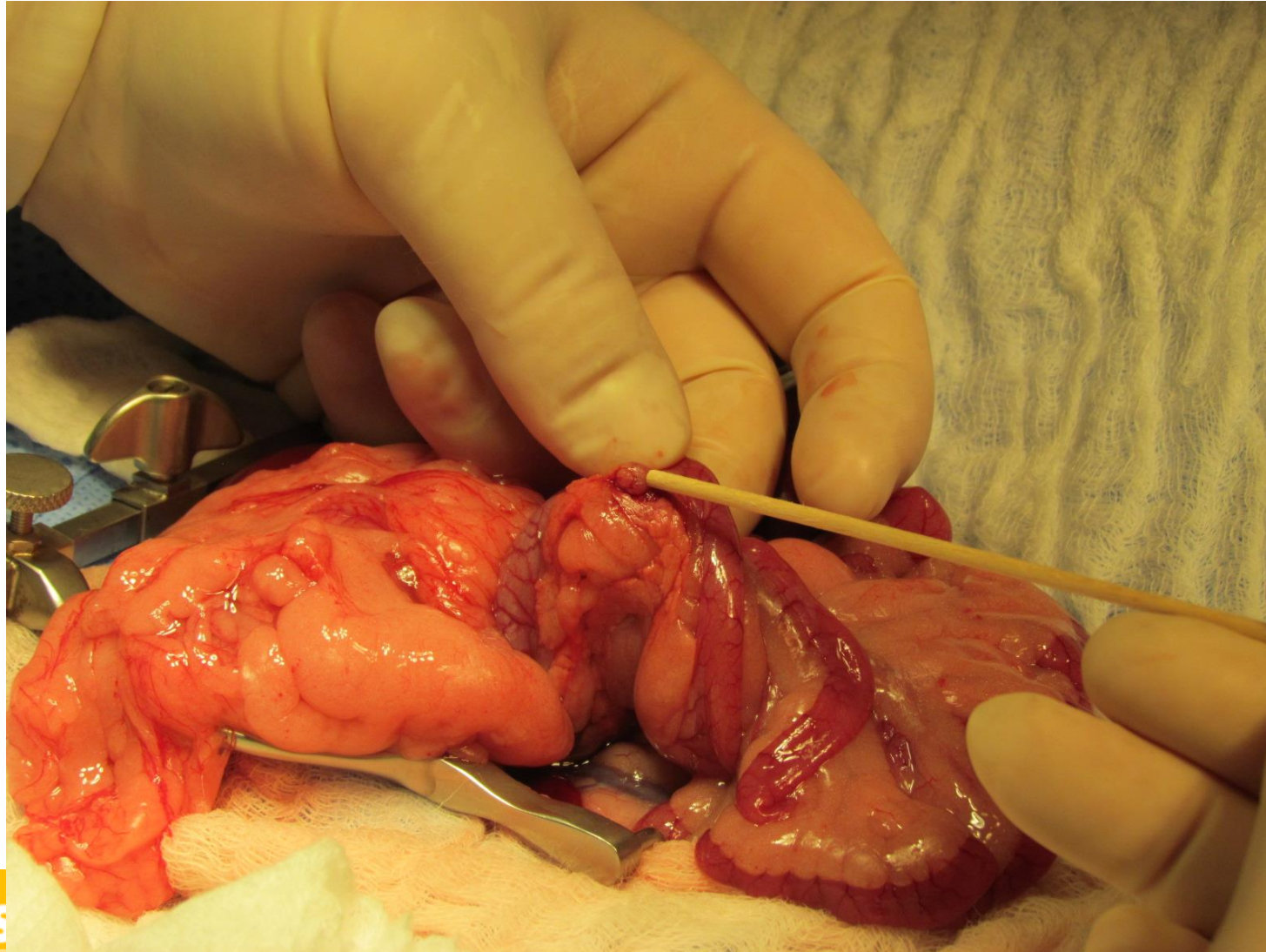
Ferret- Adrenalectomy



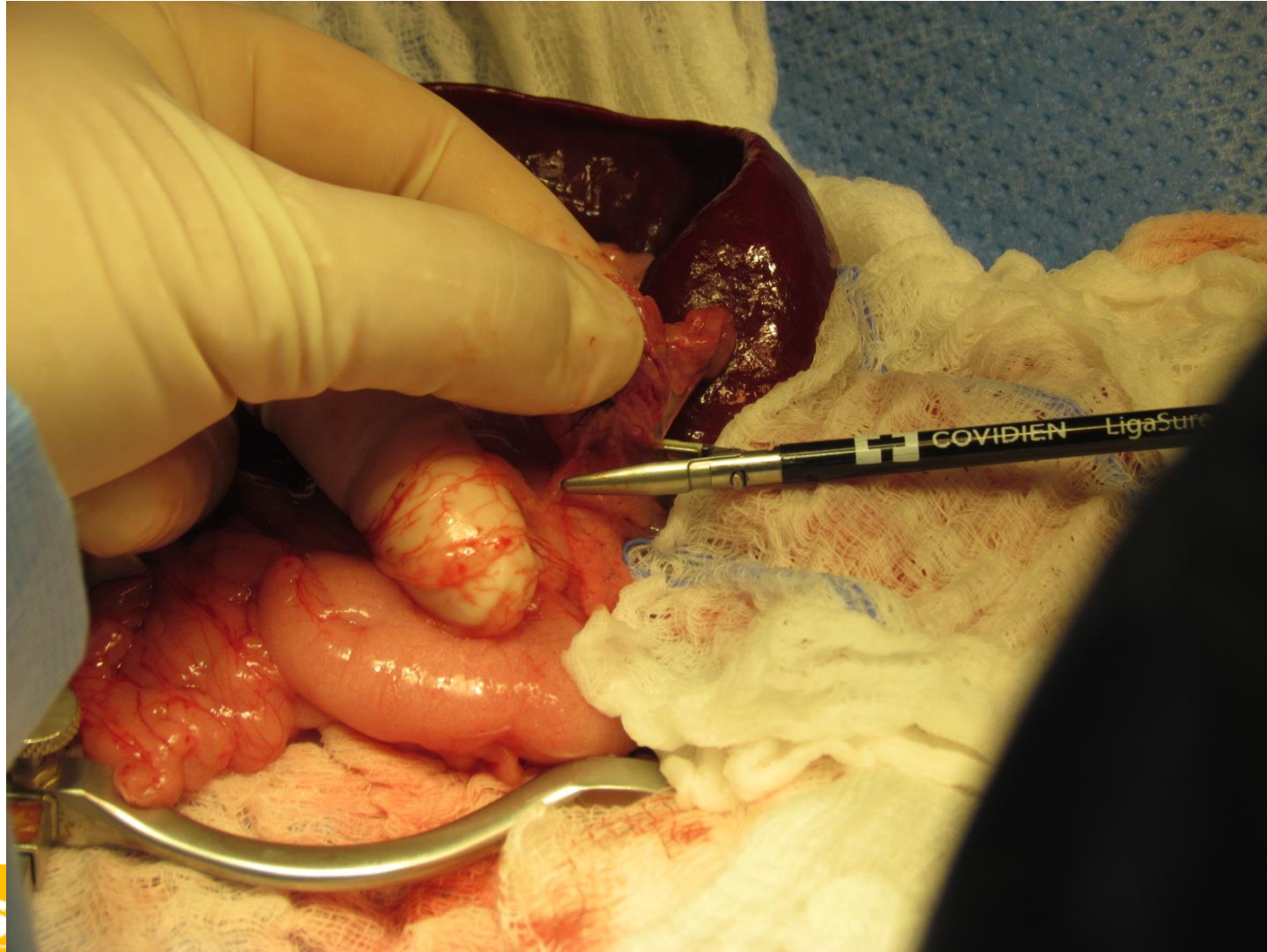
Ferret- Partial Pancreatectomy



Ferret- Partial Pancreatectomy



Ferret- Partial Pancreatectomy



Guinea Pig

- Ovariohysterectomy
- Ovariectomy
 - Ventral/dorsal approach
- Orchidectomy
 - Scrotal/prescrotal/abdominal approach
 - Open/closed technique

Guinea Pig- Orchidectomy

Scrotal Closed Technique

- Skin incision over scrotum from Cr to Ca without incising the vaginal tunic
- Undermining of the skin and exteriorize and retract the testicle and spermatic cord
- Clamp and place mass ligation x2 around spermatic cord, and transect distal to ligature
- Skin closure with subcuticular or tissue glue

Guinea Pig- Orchidectomy



Guinea Pig- Orchidectomy



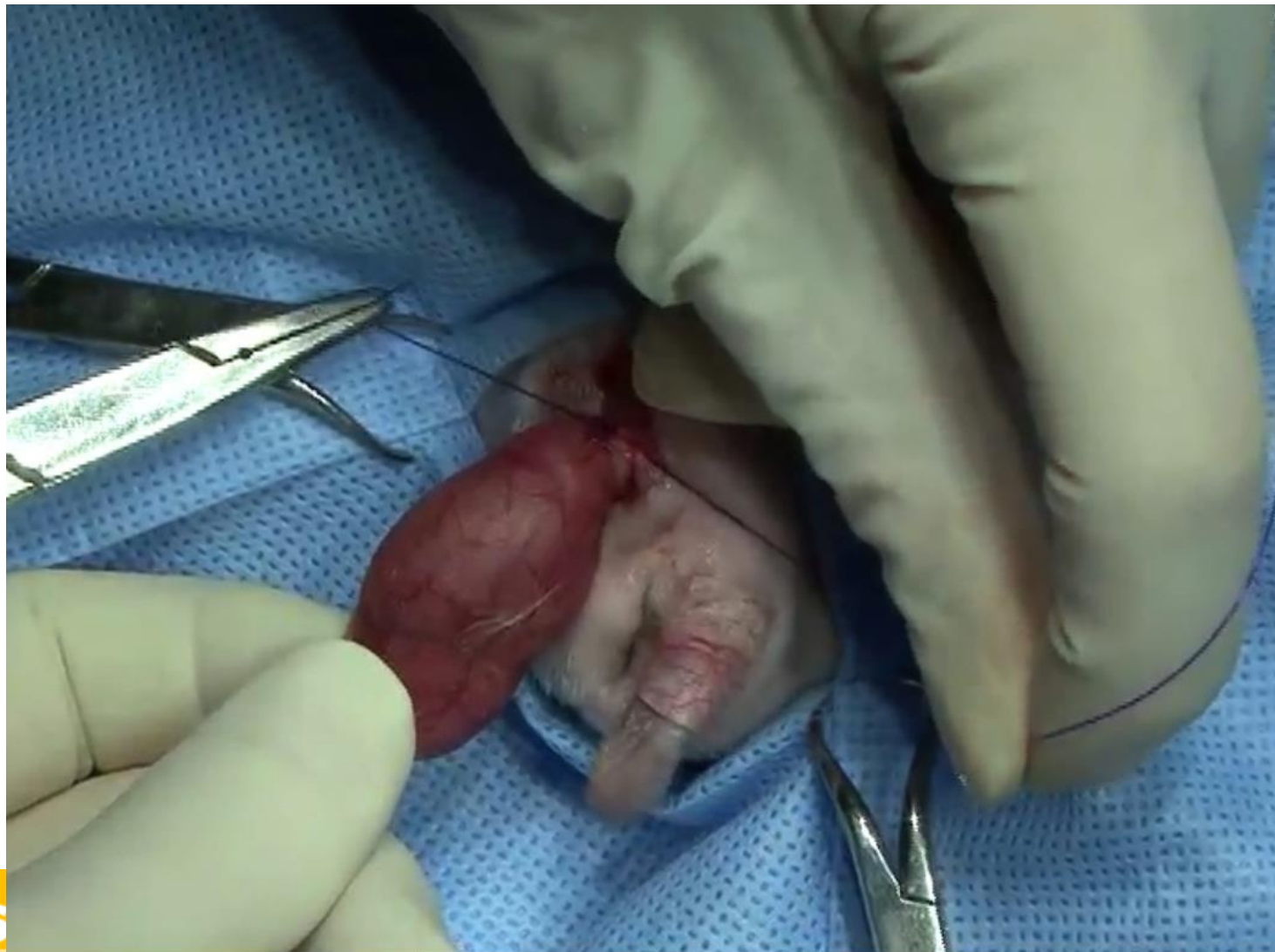
Guinea Pig- Orchidectomy



Guinea Pig- Orchidectomy



Guinea Pig- Orchidectomy



Guinea Pig- Orchidectomy



Guinea Pig- Orchidectomy



Guinea Pig- Orchidectomy

Intra-Abdominal Technique

- Skin incision midline cranial to the pubis (1-2 cm), then linea alba incision
- Push testicles into the abdomen and exteriorize them through the incision
- Separate and ligate the vascular cord and ductus deferens X 2 and transect above them
- Close fascia, SC tissue and skin (subcuticular)

Guinea Pig- Orchidectomy Intra-Abdominal Technique



Guinea Pig- Orchidectomy Intra-Abdominal Technique



Guinea Pig- Orchidectomy Intra-Abdominal Technique



Guinea Pig- Orchidectomy Intra-Abdominal Technique



Guinea Pig- Orchidectomy Intra-Abdominal Technique



Rat and Mice

- Ovariohysterectomy
- Ovariectomy
- Orchidectomy
- Mammary mass removal

Rat- Ovariectomy

- Skin and abdominal muscle incision behind the last rib and ventral to lumbar muscles
- Exteriorize mesovarium fat and ovary
- Identify the ovarian artery and vein, ligate X2 and transect above them
- Close abdominal muscle with simple interrupted and skin routinely

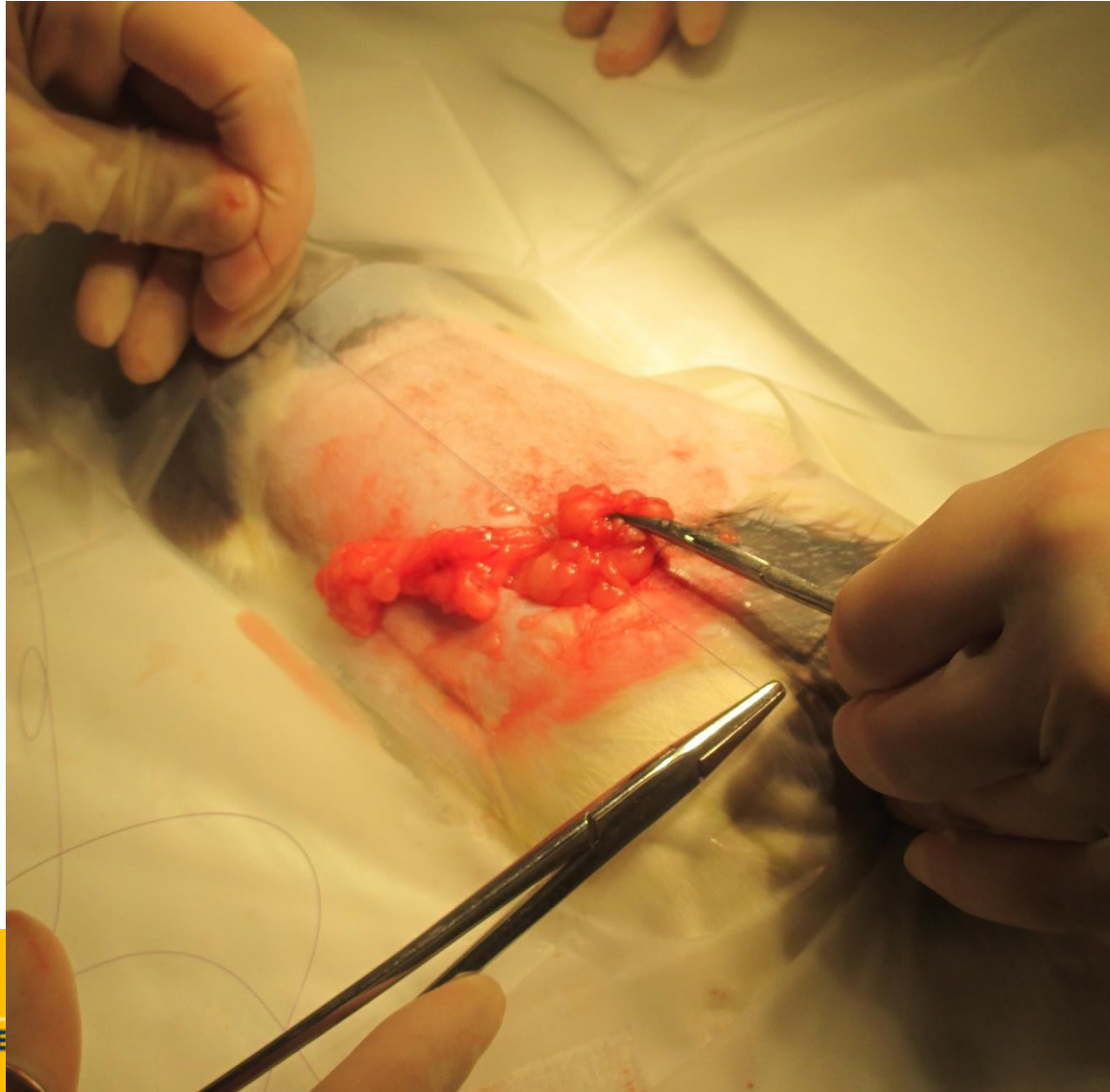
Rat- Ovariectomy



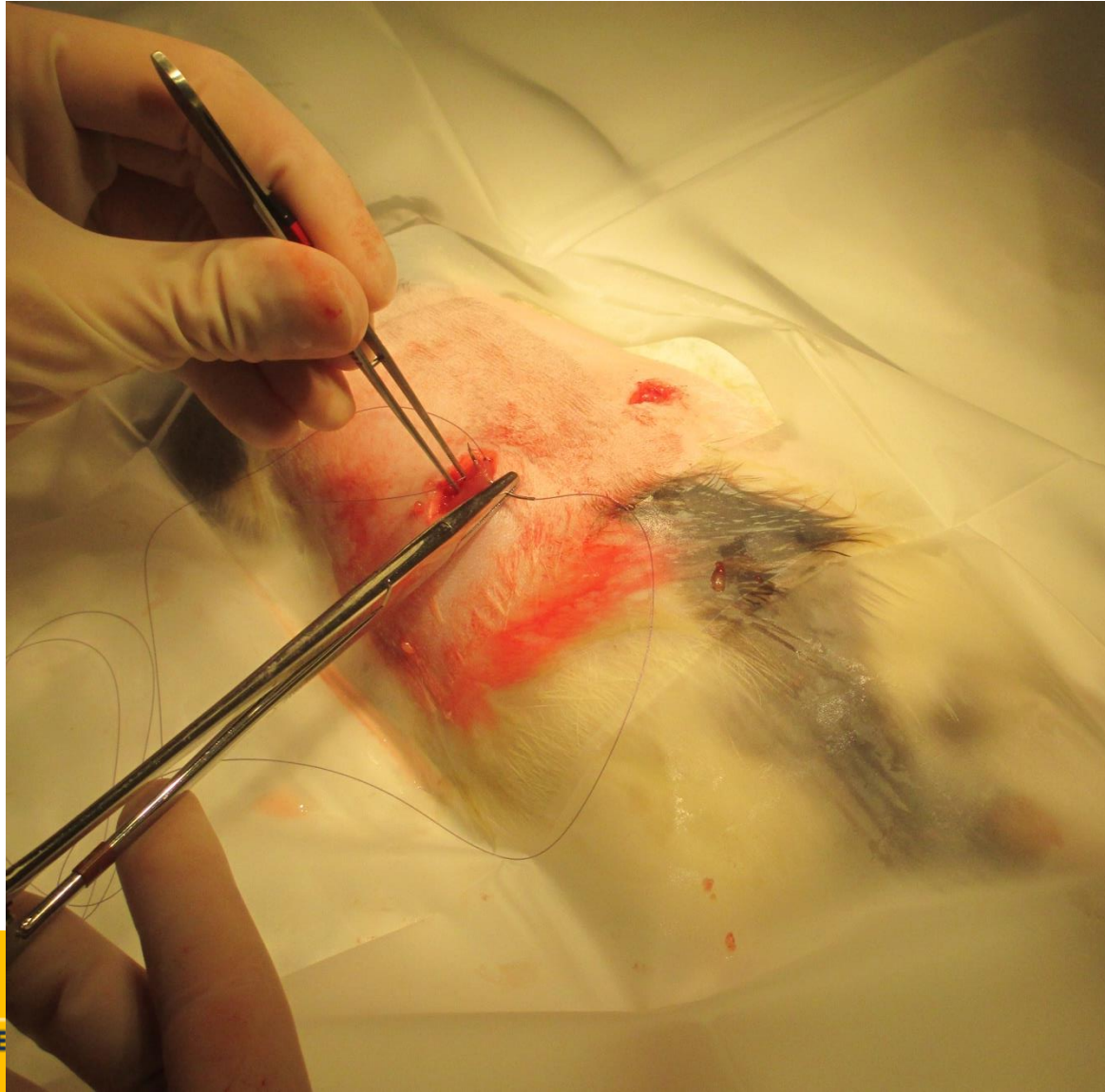
Rat- Ovariectomy



Rat- Ovariectomy



Rat- Ovariectomy



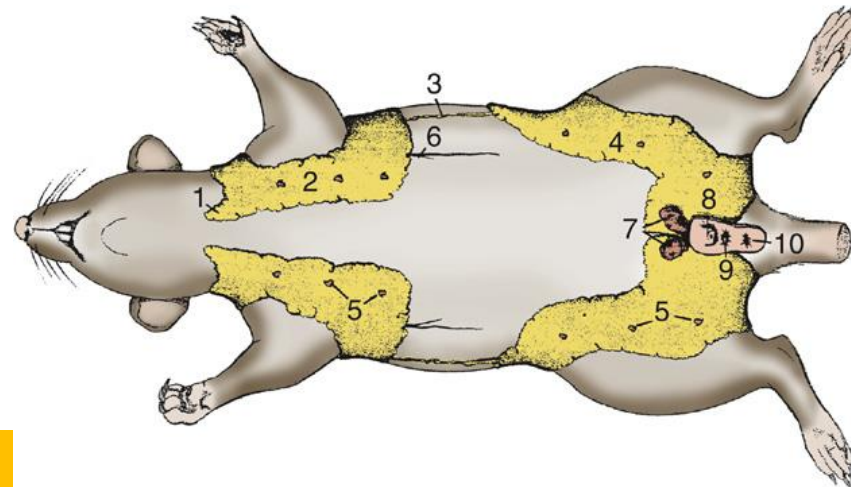
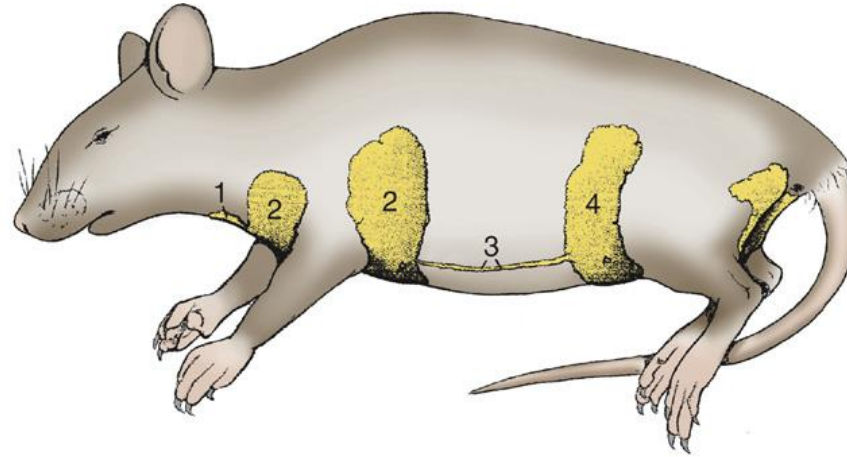
Rat- Ovariectomy



Rat- Mammary Mass Removal

- If skin ulcerated or mass suspected to be malignant, remove skin overlying tumor
- If benign save skin initially, trim later if needed
- Bluntly dissect around the mass, ligating large vessels if needed
- Close SC tissue and skin (subcuticular)

Rat- Mammary Mass Removal



Rat- Mammary Mass Removal



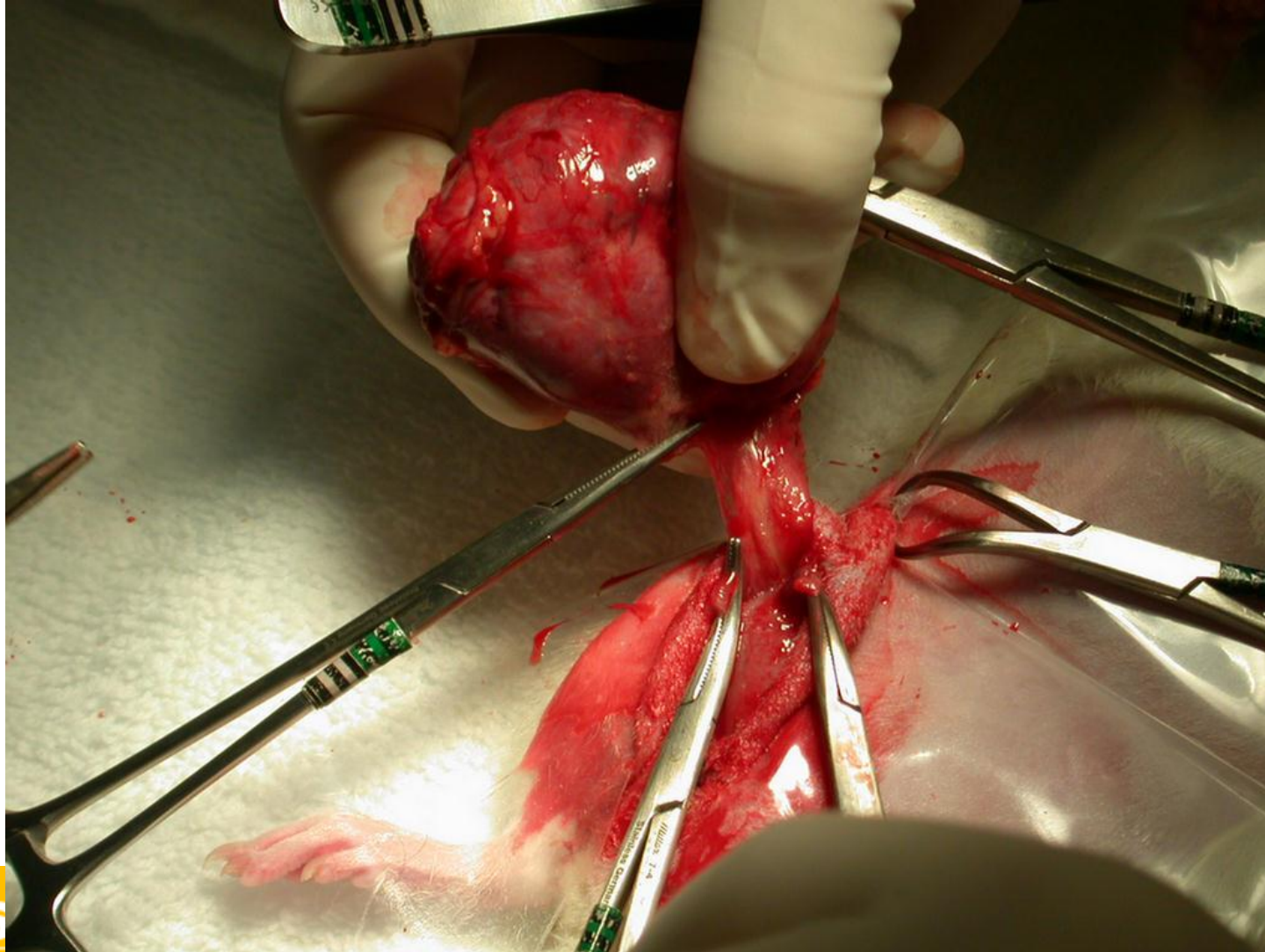
Rat- Mammary Mass Removal



Rat- Mammary Mass Removal



Rat- Mammary Mass Removal



Rat- Mammary Mass Removal



Sugar Glider

- Orchidectomy and scrotal ablation

Sugar Glider- Orchidectomy and Scrotal Ablation

- Scrotum held away from the body with gauze
- Laser tip to sever the scrotal stalk midway between the body wall and testicles

Sugar Glider- Orchidectomy and Scrotal Ablation



Thank You!

